

Six Birds Foundations VII: Lawful Theory Interaction

Ioannis Tsiokos 

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Abstract

When may two theories lawfully interact? Foundations III supplies a calculus of audited interaction within a fixed setting; this paper asks the prior question — when interaction *across* a boundary is admissible at all — and catalogs the laws that answer it. Thirty-six entries are closed, organized in four clusters — access and admission, contact and join, enablement and descent, dynamics and residue — with two questions closed as deferrals alongside them.

The central object is the *join of theories*: a composite formed from two distinct theory packages relative to a declared admissibility regime. It is not a lattice join, not a categorical product, and its existence is neither guaranteed nor free. Contact is separated from join, strictness from directionality, enablement from descent, and source independence from behavioral non-factorization; each separation is witnessed by an exhibited countermodel rather than argued.

The negative results are as load-bearing as the positive ones. Eleven scoped obstructions are established, each with named escape routes; two proposed laws are refuted, with their surviving conditional content stated; and two questions are closed as deferrals rather than left open. A join may dissolve its parents' obstructions and create new ones localized to the composite interface, so joining is not monotonically obstruction-reducing.

Provenance is stated per result, in a fixed vocabulary. Thirty-four of the thirty-six entries recover content already stated in the earlier Foundations papers or the wider corpus and sharpen it — as exact characterisations, typed records stating inherited facts together, minimized falsifiers, or scope repairs; two external results are cited for vocabulary only; one element is new, the four-way access split expressible / present / reachable / occurrent.

Grades are kept explicit and separate from verification. Fifteen entries are formal schemas, fifteen are conditional theorems with hypotheses stated inside their statements, and no entry's disposition is established by machine. The accompanying Lean development (1,477 declarations, 732 theorems, Mathlib-free) elaborates with an executed axiom receipt for every theorem: 625 depend on no axioms, 97 on `propext`, 10 additionally on `Quot.sound`, and none on `Classical.choice` or on any hole. Finite evidence is exhaustive over declared bounded carriers — 284,524 raw cases reducing to 214,636 canonical — and over nothing else.

Keywords. emergence calculus; lawful theory interaction; theory join; admissibility regime; enablement; access coordinates; bootstrap obstruction; no-go with named escapes; bounded finite carriers; audited emergence.

1 Introduction: the interaction turn

1.1 The question

Two theories are given. Each is a theory package in the sense of Foundations I: a carrier with a lens, its definability structure, a completion endomap and an audit, whose objects are the endomap’s fixed points. Each is internally lawful. The question this paper asks is what happens at the boundary between them — not how either behaves internally, and not how a single audited setting composes, but under what conditions the two may lawfully interact at all.

The question is prior to the machinery that answers it. Foundations III supplies a finite audited interaction calculus: given a declared setting, it says how records, channels and audits compose. That calculus presupposes that interaction is admissible. This paper asks when the presupposition holds, and finds that it often does not, and that the ways it fails are structured enough to be cataloged.

1.2 Why the question is not trivial

The natural first answer is that two theories interact when they share something — a carrier, an origin, an instrument, a vocabulary. Theorem 6.7 shows that this answer is wrong: common origin, common carrier and common instrument, even together, do not entail shared access; shared access does not entail source independence; and a shared lineage defeats independence-sensitive credit. Shared provenance is visible evidence of interaction and, by this theorem, inconclusive.

The natural second answer is that interaction produces a combined theory, and that the combination is some familiar construction — a product, a pushout, a common refinement. Theorem 6.4 shows that a strict join must carry an operation factoring neither through either parent nor through their declared product or common refinement, which is exactly what those constructions cannot supply. The categorical reduction is closed as a deferral (Section 12), not because it is hard but because the anti-product witness shows the identification cannot be made by fiat.

The natural third answer is that combining theories is a way of having fewer problems. Theorem 8.5 shows that a join may dissolve obstructions inherited from its parents while creating new ones localized to the composite interface. Joining is not monotonically obstruction-reducing, and the ledger distinguishes inherited from join-created residue precisely so that a composite cannot be credited with dissolving what it in fact relocated.

1.3 What goes wrong without the discipline

Each of the three wrong answers above has the same shape, and it is worth naming because the shape recurs throughout the catalog. In every case something real and observable — shared provenance, a successful combination, a dissolved obstruction — is treated as evidence for something stronger that does not follow from it. The observation is not in dispute; the inference is.

This is why so much of the paper is negative. Twenty-seven named countermodels, eleven scoped obstructions, two refutations and a discipline governing what a negative result may claim are not defensive additions to a positive theory. They are the theory: a framework for lawful interaction is mostly a framework for refusing unlawful ones, because the unlawful inferences are the available ones and they are cheap.

The discipline has a cost, and this paper pays it visibly. Every result carries its hypotheses inside its statement, every obstruction carries its escape routes, every finite figure carries its carrier, and every claim carries a grade separating what was proved from what was defined. The result is a

paper with more qualifications than assertions. That ratio is not caution; it is what the subject matter turns out to require.

1.4 What the paper establishes

Thirty-six catalog entries: thirty-four across four clusters, and the two deferrals of Section 12. Cluster A (Section 5) types what a theory can reach and on what terms something may be admitted into it, including the bootstrap obstruction: a closed admission regime with no seed and no reachable generator has no lawful first extension. Cluster B (Section 6) is the largest cluster and governs contact, join, strictness, retention, source independence, cost, and the no-free-join family. Cluster C (Section 7) types enablement against causation, descent, sufficiency and attribution, and fixes the force a negative result is permitted to carry. Cluster D (Section 8) governs iterated interaction: order residue, confluence, the arrow, and what is left over.

Twenty-one cross-family corollaries hold across cluster boundaries and are collected with their full hypothesis sets in Section 9.

What is inherited, what is sharpened, what is new. Every result carries a provenance line in the vocabulary of Section 2.4, and Section D tabulates them. Over the thirty-six entries the labels fall as follows.

Label	Entries	What it covers here
RECOVERED SBT	34	content already stated in Foundations I–Foundations VI or another corpus paper, cited with its identifier
SHARPENED	36	this paper’s contribution: exact characterisation, typed record of inherited facts, minimized falsifier, or scope repair
ELEMENTARY	3	the four quantifier forces of a negative; the monoid laws of the resource-delta fragment
RECOVERED STANDARD	2	the critical-pair criterion and Newman’s lemma (vocabulary only); limit and colimit constructions (definitions only)
NEW	1	the four-way access split expressible / present / reachable / occurrent

Nothing in this paper is presented as new that an earlier corpus paper already states; where the same separation was already exhibited there, the provenance line says so and says whether this paper’s version is identical, stronger, weaker or differently scoped. The corollaries are typed records stating inherited facts together; the countermodels are minimized falsifiers, twenty-two of which recover separations the corpus already exhibits.

1.5 What the paper does not establish

The catalog is not claimed complete: the status calculus of Definition 6.3 is exhaustive over its declared interface family and is not asserted to be so outside it. No instantiation is carried out; Section 11 indicates where the laws are expected to bite and states what each application would have to supply. Two candidates are refuted rather than established, and two are closed as deferrals.

Nothing in this paper is proved by machine in the sense of having its mathematical grade established there. The Lean development confirms that the recorded statements elaborate and depend on exactly the axioms reported in Section 14.2; it cannot confirm that they are the right statements. Section 13 collects the boundaries in one place.

1.6 Relation to external frameworks

Several mature literatures address neighboring, more specialized problems. Nelson–Oppen combination gives conditions under which decision procedures for quantifier-free fragments of suitably separated first-order theories cooperate; institution theory and algebraic specification study signatures, sentences, models, satisfaction, morphisms, and structured specifications across logical systems [20, 12, 25]. Interface automata, compositional specifications, and contract-based design make compatibility, refinement, environment assumptions, and composition explicit [10, 1, 5]. These are the nearest technical comparison classes for the present work, but none is invoked as a representation theorem for the regime-relative join defined here.

Theories can also be related without being composed. Morita and categorical equivalence provide criteria for comparing formulations and expose how much interpretive structure an equivalence criterion preserves [3, 41]. They motivate the categorical deferral in Section 12; they do not imply that a VII join exists or reduces to such an equivalence.

The operational side touches classical distinctions among safety, liveness, reachability, and bounded state-space search. Alpern and Schneider supply the safety/liveness distinction, while model-checking treatments supply the reachability and bounded-search comparisons [2, 8, 7]. The seven access coordinates and five-stage operational spine used here are not claimed to be standard reformulations of those notions; the external literature supplies comparison points and, especially, the warning that a bounded search result remains bounded.

Provenance, causation, and enablement supply three further neighboring vocabularies. Database provenance tracks how outputs depend on source data; causal models separate common-cause structure from stronger causal claims; organizational-closure work treats constraints as acting on processes while depending on and contributing to one another’s maintenance [13, 23, 19]. VII keeps these tasks separate: its source gate is not statistical independence, and its enablement record is not identified with a causal or biological closure model.

Finally, Berry’s cyclic adiabatic geometric phase and Crooks’s fluctuation theorem provide domain-specific comparisons for circuit dependence and forward/reverse asymmetry [6, 9]. Lean’s architecture and Pollack’s trust analysis distinguish kernel checking of encoded derivations from broader confidence in the formalization and checking process [11, 24]. These sources are used below to police analogies and validation claims, not to import physical dynamics or establish conceptual adequacy.

1.7 Relation to the rest of the series

Foundations I supplies the closure mechanics this paper assumes throughout: a theory package is a carrier with a lens, a definability structure, a completion endomap and an audit, its objects are the fixed points of the endomap, and the primitives P1–P6 construct layer formation. Foundations II supplies the admissibility vocabulary that Cluster A specialises. Foundations III supplies the audited interaction calculus this paper catalogs over, and the distinction between the two was stated above. Foundations IV supplies the catalog form itself and several inherited terms — access quotient (Def. 2.5), no-free-distinction (F12), adequacy and no-overread (F11) — used here unchanged.

The relationship to Foundations V and Foundations VI is closer and correspondingly more dangerous. Both are *vendored*: their formal libraries are imported byte-identically and built upon. That makes their vocabulary immediately available and immediately liable to be misread, and Section 4 is the section that manages it. One collision in particular — the word *join*, which Foundations V uses for a lattice operation on a common carrier — would silently corrupt every result in Cluster B if left unmarked.

1.8 How to read the catalog

Each entry carries a disposition — what it is — and a verification label — whether its formal counterpart elaborates. The two are independent and never merge (Remark 2.1). Every entry states its hypotheses inside its statement, its nonclaims beside it, and, where it is an obstruction, its escape routes. Every finite figure travels with the carrier it was computed over.

Section 2 fixes the conventions, Section 3 the six definitions the catalog rests on, and Section 4 what is inherited from Foundations I–Foundations VI and what must not be carried over.

2 Apparatus and conventions

This section fixes what is assumed, what is defined here, and how results are graded. It introduces no laws. A reader who knows Foundations I–Foundations VI may skim Section 2.1 and should not skip Section 2.3, which governs how every later statement is to be read.

2.1 What is assumed

We assume the closure mechanics of Foundations I [30] and do not restate them: a theory package is a carrier with a declared lens, its definability structure, a packaging or completion endomap and an audit (the finite theory package of Foundations I Def. 1 and §3.4), its objects are the fixed points of that endomap, and the primitives P1–P6 construct layer formation rather than merely describing it. We assume Foundations II’s admissibility vocabulary [31] (admissible claim under honest bookkeeping, Def. 3.1), of which the admission layer below is a specialisation, and Foundations III’s finite audited interaction calculus [32], over which this paper is a catalog.

The relationship to Foundations III is worth stating precisely, because the two titles are close enough to mislead. Foundations III supplies a calculus of audited interaction *within* a declared setting: how records, channels and audits compose when the setting is fixed. This paper asks a different question — when is interaction *between two theories* lawful at all — and catalogs the laws that answer it. Foundations III is the machinery; Foundations VII is the legislation about when the machinery may be applied across a boundary. Neither subsumes the other, and results are not transported between them without an explicit bridge (Section 4).

We also inherit, and do not re-derive, the vendored formal libraries of Foundations V [34] and Foundations VI [35]. These are imported byte-identically; nothing in this paper edits them, and the import is audited rather than assumed (Section 4).

2.2 Theory packages, regimes, audits

A *theory package* $\mathcal{T}$ is the Foundations I object: carrier, declared lens, induced definability structure, packaging endomap, and audit. We write $\mathcal{T}_1, \mathcal{T}_2$ for the two packages whose interaction is at issue.

An *admissibility regime* $\mathcal{R}$ is the declared standard against which an interaction is judged: which operations are lawful, which sources may be drawn on, which costs must be paid, and what the audit must record. Nothing in this paper is asserted absolutely; every existence, obstruction and cost claim is relative to a declared $\mathcal{R}$. Where a regime is left implicit in a statement, the statement holds for every regime satisfying its stated hypotheses.

An *audit* $\mathcal{A}$ is an append-only record. Append-only is load-bearing rather than decorative: several results below turn on the fact that a failed, revoked or withdrawn entry survives in the record instead of disappearing, so that a later reader can see that something was tried and refused.

2.3 How results are graded

Every result in this paper carries two independent labels. Conflating them is the most likely way to misread the paper, so they are stated separately and never merged.

The **disposition** says what a result *is*. Six dispositions occur:

- **CONDITIONAL THEOREM** — a theorem whose hypotheses are stated inside its own statement. It holds under those hypotheses and is asserted nowhere else.
- **FORMAL SCHEMA** — a definitional or typed schema. It fixes an object and what may be recorded about it. It is not a theorem and proves nothing.
- **REFUTED CANDIDATE** — an unconditional form shown false. What survives conditionally is stated alongside it, never omitted.
- **CLOSED DEFERRAL** — a question closed by declaring it out of scope, with the landed fragment named. A deferral is not an open problem awaiting work in a later paper; it is a boundary drawn deliberately.
- **CONSTRUCTIVE COUNTERMODEL** — an exhibited witness refuting a named claim.
- **LEAN DECIDABLE FINITE** — a property decidable over a declared finite carrier, always quoted with that carrier’s cardinality.

The **verification** label says whether the recorded formal assets elaborate in the Lean kernel and whether every theorem among them carries an executed `#print axioms` receipt [17]. Every asset in this paper is kernel-verified in that sense (Section 14).

Remark 2.1 (The two axes do not interact). Verification does not upgrade a disposition. A **FORMAL SCHEMA** that elaborates in Lean is a schema whose formalization compiles; it has not become a theorem. This paper contains *no* result whose disposition is “proved by the kernel”: the kernel confirms that the recorded declarations are well formed, that their proofs are complete, and that they depend on exactly the axioms we say they depend on. It cannot confirm that a schema is a theorem, because that is not a question a proof assistant is asked.

Remark 2.1 is the reason phrases such as “machine-verified theorem” do not appear in this paper. Where we say *proved*, we mean a **CONDITIONAL THEOREM** with its hypotheses; where we say *typed*, *recorded* or *defined*, we mean a **FORMAL SCHEMA**. The distinction is also checked mechanically: a drafting lint pass scans for grade-inappropriate verbs attached to a named result, for forbidden verification phrases, and for unqualified scope words. That pass is a heuristic aid, not a proof of compliance — it can be satisfied by a carefully worded overclaim — so it supplements review rather than replacing it.

2.4 Provenance vocabulary

Every theorem, definition, refuted remark, deferral, no-go front, corollary and countermodel in this paper carries a provenance line saying what is inherited and what is this paper’s. Five labels are used, each with a clause naming what it covers. **RECOVERED STANDARD**: a known result of the external literature, cited from a verified primary source. **RECOVERED SBT**: already stated in an earlier paper of the Six Birds corpus, cited with that paper’s own identifier. **ELEMENTARY**: an identified elementary fact with no canonical source. **SHARPENED**: this paper’s own contribution in one of four forms — an exact characterisation, a typed record stating inherited facts together, a minimized falsifier, or a scope repair. **NEW**: present in no cited source, used only where a duplication search against every earlier corpus paper and the external anchors supports it. A result may carry several labels. Each line also says, in one sentence, whether this paper’s version is identical to, stronger than, weaker than or differently scoped from its canonical source, and names the Lean

declarations that carry the statement together with their executed axiom receipt. Section D collects the labels in one table.

2.5 Finite evidence and its boundary

Many results below are accompanied by exhaustive enumeration over a bounded carrier. The convention is absolute: *a finite figure never travels without its carrier*. “Exhaustive” in this paper always means exhaustive over a declared carrier of stated size, never exhaustive simpliciter.

Summed over all 43 assays the carriers comprise 284,524 raw cases, reducing to 214,636 canonical cases, of which 70,915 are accepted and 143,721 rejected under the declared laws. The eleven Phase-5 closure assays together account for 86,912 raw and 84,864 canonical (21,081 accepted, 63,783 rejected); that aggregate is the census the release validation checks directly, and is not the same as the all-assay total above. The largest *single* assay is the strictness family at 131,072 raw, 65,536 canonical, 32,770 accepted. Individual assays name their own carriers where they are used.

What exhaustive enumeration buys is a guarantee that no case *inside the declared carrier* was overlooked. What it does not buy is any statement about cases outside it. Where a law is stated for all theories satisfying certain hypotheses, the warrant is the proof, not the enumeration; the enumeration is a control against the proof having quietly assumed something the hypotheses do not supply.

2.6 Notation

We write $\mathcal{T}_1 \bowtie_{\mathcal{R}} \mathcal{T}_2$ for the join of theory packages $\mathcal{T}_1$ and $\mathcal{T}_2$ under regime $\mathcal{R}$. The symbol $\bowtie$ is chosen deliberately and is not a lattice operation; Section 4.3 explains what it is not, and why the more obvious symbol is unavailable.

Identifiers of the form VII-C001, VII-001, FT01, NGVII-01 and FVII-COR-001 are locators into the accompanying formal artefacts. They appear in tables and parentheticals so that any statement can be traced to its recorded source. They carry no argumentative weight: nothing in this paper is true because it has an identifier.

3 The definitional layer

Six definitions carry the paper. They are stated once here and used without restatement thereafter. Each is a FORMAL SCHEMA: it fixes an object and what may be recorded about it, and proves nothing. The laws that follow in Sections 5 to 8 are statements *about* these objects.

3.1 D1 — Access coordinates

Definition 3.1 (Operational access record). An *access record* for a theory package assigns seven separately recorded Boolean coordinates: whether a content is **expressible**, **present**, **exposed**, **recoverable**, **admissible**, **reachable**, and **occurent**. The record is *coherent* when it satisfies the admitted implication spine

$$\text{occurent} \Rightarrow \text{reachable} \Rightarrow \text{admissible} \Rightarrow \text{present} \Rightarrow \text{expressible}, \quad \text{recoverable} \Rightarrow \text{exposed} \Rightarrow \text{present},$$

and carries a non-empty audit.

Provenance. RECOVERED SBT (exposed / recoverable / admissible). NEW (expressible / present / reachable / occurent). SHARPENED (the coherent seven-coordinate record). *Canonical:* Foundations IV Def. 2.5, F10–F12

(Thms. 3.2–3.4) [33]; Foundations II Def. 3.1 [31]. *Relation*: Same as VII-C001. *Lean*: `FoundationsVII.DomainState, FoundationsVII.DomainState.Coherent, FoundationsVII.DomainState.normalForm_eq`; receipt axioms: none.

The content of Definition 3.1 is the *separation*: seven coordinates, not one notion of availability wearing seven hats. The spine is stated in one direction only. No converse is assumed, and Section 5 exhibits a coherent witness for each non-implication, so the separations are not merely unproven but refuted as implications.

Remark 3.2 (Why seven). The four-way split *expressible/present/reachable/occurrent* is this paper’s own; *exposed*, *recoverable* and *admissible* specialise inherited status machinery. The seven are not a taxonomy for its own sake: each is the antecedent or consequent of some law below, and dropping any one would collapse a distinction a later result depends on.

Which collapses the spine forbids. Six non-implications are load-bearing, and each has a named witness in the reference world. *expressible* does not give *present*: a theory can say something that is not there. *present* does not give *exposed*: content can be there and not visible at the interface. *exposed* does not give *recoverable*: visible is not the same as retrievable, and Definition 5.2 names the overread that treats it so. *admissible* does not give *reachable*: a permitted transition may have no route to it (CM-08). *reachable* does not give *occurrent*: a route existing is not a route taken (CM-09). And *recoverable* does not give *admissible*: getting content back is not the same as being permitted to use it.

The two chains are also independent of each other. The *occurrent*→*expressible* chain runs through *admissibility* and *presence*; the *recoverable*→*present* chain runs through *exposure*. They meet only at *present*, which is why *exposure* can be absent in a record whose occurrence chain is fully satisfied.

3.2 D2 — Admissibility regime

Definition 3.3 (Admissibility regime). An *admissibility regime* $\mathcal{R}$ declares: the operations lawful in an interaction, the sources that may be drawn upon, the costs chargeable, and the audit obligations. Every existence, obstruction, and cost claim in this paper is relative to a declared $\mathcal{R}$.

Provenance. RECOVERED SBT (admissibility as a declared standard). SHARPENED (regime-relativity of every existence, obstruction and cost claim). *Canonical*: Foundations II admissibility meta-theory [31]. *Relation*: Differently scoped: Foundations II judges claims; VII judges interactions. *Lean*: `FoundationsVII.AdmissionRegime, FoundationsVII.AdmissionRegime.open_family_is_not_covered_by_closed_bootstrap_hypotheses`; receipt axioms: none.

Regime-relativity is not a hedge. It is what makes the obstructions in this paper falsifiable: a no-go stated relative to $\mathcal{R}$ is refuted by exhibiting a lawful $\mathcal{R}$ in which the obstruction fails, and several of the escape routes below are exactly that — NGVII-01’s open-external-provision escape is a regime in which the family is not closed, and it is a lawful regime rather than a loophole.

The corresponding limit is that no result here transfers between regimes for free. A law established under $\mathcal{R}_1$ says nothing about $\mathcal{R}_2$ unless a bridge records what the change of regime preserves and loses. This is the single most likely way a result from this paper would be misapplied: quoting an obstruction established under a restrictive regime as though it held under a permissive one.

3.3 D3 — Contact surface

Definition 3.4 (Contact surface). A *contact surface* between $\mathcal{T}_1$ and $\mathcal{T}_2$ records that an interaction occurred: the participating packages, the mode of contact, the channel evidence licensing that mode, and the audit. Contact is *witnessed* when the channel evidence licenses the declared mode.

Provenance. RECOVERED SBT (instrument-indexed records and visibility bridges). SHARPENED (contact as a witnessed surface with five licensed modes). *Canonical:* Foundations III §11 [32]. *Relation:* Differently scoped: VII’s surface records contact between two packages. *Lean:* FoundationsVII.ContactSurface, FoundationsVII.ContactWitness, FoundationsVII.ContactMode, FoundationsVII.ContactChannelEvidence.rendezvous_licensed; receipt axioms: none.

Contact is deliberately weaker than join. Two theories may touch — share a carrier, exchange a record, transport a payload — without any joint object coming into existence. Section 6 separates the two, and the separation is one of the paper’s load-bearing negative results.

The motivation resembles interface automata’s distinct treatment of compatibility and composition and contract-based design’s explicit treatment of assumptions, guarantees, refinement, and composition [10, 5]; this is a comparison of roles, not an identification of formalisms.

Modes and their evidence. The mode of contact is not free: each mode has its own licensing condition, and declaring a mode without the matching evidence is not contact but a claim of contact. Rendezvous requires a direct endpoint match; mediation requires a certified mediator; sequential contact requires a recorded sequence; shared-carrier contact requires a certified shared carrier; shared-budget contact requires a certified shared budget. The five are not interchangeable, and a system exhibiting one does not thereby exhibit another — CM-19 is exactly a shared carrier that licenses no transport.

3.4 D4 — Join of theories

Definition 3.5 (Join of theories). For theory packages $\mathcal{T}_1, \mathcal{T}_2$ and admissibility regime $\mathcal{R}$, a *join* $\mathcal{T}_1 \bowtie_{\mathcal{R}} \mathcal{T}_2$ is a package that (i) is compatible with both parents in the sense recorded by the join entry, (ii) is admissible under $\mathcal{R}$, and (iii) has its formation cost discharged in the budget ledger. A join is *strict* when it is not obtainable by relabelling, rescheduling, coarsening, or common refinement of its parents.

Provenance. RECOVERED SBT (the repair join, cited only to be excluded). SHARPENED (a conditional, costed join of two packages with strictness defined by exclusion of four cheap constructions). *Canonical:* Foundations V D1 repair join (Def. 3.1) [34]; Foundations IV F8 (Lemma 4.3), F51 (Thm. 8.18) [33]. *Relation:* Not the repair join (Remark 4.1); differently scoped from F51. *Lean:* FoundationsVII.JoinCandidate, FoundationsVII.JoinCertificate, FoundationsVII.StrictJoinEvidence.Certified, FoundationsVII.StrictJoinEvidence.certified_has_objecthood; receipt axioms: none.

Three features of Definition 3.5 are worth flagging immediately, because each is a place where an inherited intuition would mislead. Existence is *conditional*: nothing in the definition asserts that a join exists. Formation is *costed*: clause (iii) is not bookkeeping, and Section 6 shows that dropping it admits free joins that the framework must refuse. And strictness is *defined negatively*, by exclusion of four specific cheap constructions — because the interesting failure mode is not the absence of a joint object but the presence of a fake one.

Neighboring frameworks pose different problems and side conditions: Nelson–Oppen combines decision procedures for quantifier-free fragments under separation and other hypotheses; institution theory and algebraic specification organize signatures, models, satisfaction, morphisms, and structured specifications; contract frameworks define their own composition and refinement operators [20, 12, 25, 5]. They are comparison baselines for the VII entry and bridge records, not interchangeable meanings of *join*.

Per Remark 4.1, $\bowtie$ is not Foundations V’s repair join and shares none of its lattice properties.

3.5 D5 — Enablement

Definition 3.6 (Enablement). $\mathcal{T}_1$ *enables* $\mathcal{T}_2$ when $\mathcal{T}_1$'s formation is a recorded precondition of $\mathcal{T}_2$'s, with typed attribution: the enabling source, whether execution was witnessed, whether the budget was accounted, whether the audit records it, and whether any executor was hidden.

Provenance. RECOVERED SBT (enablement versus within-layer causation). SHARPENED (a typed relation with attribution and hidden-executor fields). *Canonical:* §1.3, §11.2 [27]; Foundations V E10 [34]. *Relation:* Stronger: the corpus distinction, typed. *Lean:* FoundationsVII.EnablementRecord, FoundationsVII.AttributedEnablement, FoundationsVII.AttributedEnablement.wellFormed_has_no_hidden_executor; receipt axioms: none.

Enablement is typed against four confusions, each of which is separated by an explicit result in Section 7: it is not descent, not sufficiency, not within-layer causation, and not an attribution to whoever happened to be describing the system. The last is the practically dangerous one — a theorist who notices a layer is not thereby its enabler — and the framework refuses attribution credit when the executor is hidden.

3.6 D6 — Residual

Definition 3.7 (Residual). A *residual* is what an interaction leaves behind that its declared ledgers do not discharge: unpaid cost, undissolved obstruction, or unattributed enablement. A residual is recorded in an append-only ledger and is either discharged, carried, or declared.

Provenance. RECOVERED SBT (typed residual status; adequacy residual). SHARPENED (discharged / carried / declared as the only three dispositions of a residual). *Canonical:* Foundations IV F9 (Thm. 4.4) [33]; Def. 4.2 [26]. *Relation:* Differently scoped: VII's residual is what an interaction leaves undischarged. *Lean:* FoundationsVII.ResidualEntry, FoundationsVII.ResidualLedger, FoundationsVII.ResidualLedger.append_extends; receipt axioms: none.

The three options partition the possibilities by construction, and that is the point. There is no fourth option in which a residual is quietly dropped, and Section 8 turns this into a law about what interaction dynamics can and cannot dispose of.

4 Inheritance and bridge discipline

This paper stands on six predecessors and, through two vendored scaffold trees, builds on formal libraries drawn from across the series. Inheritance on that scale is a liability as much as an asset: a term reused in a shifted sense, or a result transported across a boundary it was never proved to cross, would compromise everything downstream. This section states what is inherited, what is renamed, and what must not be carried over.

4.1 What is vendored

The Lean development imports the Foundations V and Foundations VI libraries directly, together with the Foundations I [30] closure-ladder core, the Foundations II [31] admissibility library, the Foundations III [32] interaction library, and the shared meta-mathematics library. These trees are included byte-identically and are treated as immutable: no result in this paper is obtained by editing an inherited file. That immutability is checked mechanically rather than promised, and the check is part of the release validation (Section 15).

Name collision across so large an import is a real risk and was measured rather than assumed. Against roughly 2,400 declaration names in the vendored scaffolds (the exact count depends on

which declaration keywords the scan admits), the 1,352 names this paper introduces collide in exactly one case (**Stable**, which also occurs in Foundations VI). Namespacing separates them in the formal development. In prose, where there are no namespaces, the term is qualified on first use.

4.2 Terms inherited unchanged

The following are used in exactly their established senses and are not redefined here: *theory package*, *carrier*, *lens*, *definability structure*, *audit*, *record*, *status family*, *descent obstruction*, *access quotient*, *no-free-distinction*, and *adequacy/no-overread*. Where this paper appears to say something new about one of them, it is saying something new about *interaction* between theories that use them, not revising the term.

4.3 The join collision

One inherited term requires an explicit stop, because the collision is silent and the wrong reading is the natural one.

Foundations V [34] defines, as its Definition D1 (Def. 3.1), the *repair join*. Quotients on a common carrier are ordered by refinement, and for quotients Q, R the repair join $Q \vee R$ is the coarsest quotient refining both. It is a lattice join: the carrier is shared, the operation is an upper bound, and it always exists.

The object this paper calls a *join of theories* is none of those things.

Remark 4.1 (What $\bowtie$ is not). Three differences from Foundations V’s repair join, each of which independently breaks the lattice reading:

1. It relates two *distinct theory packages* under a declared admissibility regime, not two quotients of one carrier. (It does not require that the two packages have *different* carriers; shared-carrier interaction is squarely in scope, and one of the countermodels below is exactly a shared carrier without lawful transport. What differs is that the objects joined are packages, not quotients.)
2. It is not a lattice upper bound. There is no ambient order in which the join is a least element above both arguments.
3. Its existence is *conditional and costed*. The repair join is a total operation on its lattice; a join of theories may fail to exist, and where it exists it is paid for.

The two constructions therefore get different symbols. $\vee$ remains Foundations V’s; this paper writes $\bowtie$ and never abbreviates the join of theories to bare “join” at the first use in a section. The symbol $\bowtie$ was selected because it occurs nowhere in the existing corpus and so carries no inherited meaning, while remaining suggestive of joining to a reader who meets it cold.

Two results in Section 6 are stated at the point of definition rather than deferred, precisely so that the lattice reading cannot form and then have to be dislodged: that a join need not exist, and that it is not a categorical product.

4.4 Enablement: a promotion, not a clash

Foundations V uses the word *enablement* four times and never defines it, importing the enablement-versus-within-layer-causation distinction from the agenthood literature [27]. This paper promotes the term: enablement becomes a first-class typed relation with its own attribution calculus (Section 7).

The promotion is declared rather than silent, and the term is typed against the four things it is most often confused with — causation, descent, sufficiency, and attribution to theorist, carrier or observer. Where this paper says *enablement*, it means the VII relation, and readers arriving from Foundations V should not assume the earlier informal usage was already this.

4.5 Bridge discipline

Nothing is transported from an inherited theory into a VII statement except through a recorded bridge. A bridge names the source declaration, the target declaration, the map between them, the hypotheses it preserves, the hypotheses it adds, and the hypotheses it loses. A bridge that loses a hypothesis is not thereby illegitimate — it is legitimate exactly when the loss is recorded and the downstream claim is weakened accordingly.

The discipline exists because the failure it prevents is invisible. A result imported without its hypotheses looks identical to a result proved here, and the difference surfaces only when someone tries to use it outside the range where the import was valid. Recording the bridge makes the boundary checkable rather than remembered.

Citation alone does not license transport. A citation records where something was said; a bridge records that it may be used here. This paper’s formal development distinguishes the two explicitly and treats an uncertified bridge as an unlicensed transport regardless of how well-attested the source is.

5 Cluster A — Access and admission

Nine entries govern what a theory can reach, on what terms something may be admitted into it, and how claims about either may be made at all — six substantive entries and the three protocol-discipline entries of Section 5.5. The cluster is prior to interaction: before two theories can lawfully meet, each must have a well-typed account of its own access and its own admission discipline. The separations established here are used throughout Sections 6 to 8.

5.1 The access record

Definition 5.1 (Accessible-domain normal form, FORMAL SCHEMA, VII-C001). A theory package admits a typed operational-domain record separating the seven coordinates of Definition 3.1. The record is in normal form when each coordinate is independently determined and the implication spine holds.

Provenance. RECOVERED SBT (the exposed, recoverable and admissible coordinates). NEW (the four-way split expressible / present / reachable / occurrent). SHARPENED (the seven-coordinate coherent record with a one-directional spine and a coherent witness for every converse). *Canonical:* Foundations IV Def. 2.5 (access quotient), F10 (Thm. 3.2), F11 (Thm. 3.3) [33]; Foundations II Def. 3.1 (admissible claim under honest bookkeeping) [31]; Lean adapters FT01: `SixBirds.ClaimRecord`, `SixBirdsFoundationsV.AccessPolicy`. *Relation:* Differently scoped: the inherited coordinates are quotient- and claim-level notions; VII records them as Boolean coordinates of one typed record and adds a four-way coordinate split that no earlier corpus paper states as a tuple — the words *expressible* and *reachable* occur earlier in other roles, *occurrent* does not (duplication search: `paper_prep/PP-03` §2, confirmed in review). *Lean:* `FoundationsVII.DomainState`, `FoundationsVII.DomainState.Coherent`, `FoundationsVII.DomainState.normalForm_eq`; receipt axioms: none.

Positive normal form. The separation is the content. A single notion of “availability” collapses seven distinct questions — can it be said, is it there, is it visible, can it be got back, is it permitted, can it be got to, did it happen — into one, and every collapse licenses an inference the framework must refuse.

Strictness. Each implication in the spine is one-directional, and each non-implication has a coherent witness. The witnesses are not hypothetical: for every arrow in Definition 3.1 the finite battery contains a coherent record satisfying the antecedent’s negation together with the consequent, so the converses are refuted rather than merely unproven.

Converse and partition. Domain width is not capability. A theory whose record has many coordinates set is not thereby more able; the coordinates record status, not power.

Exact finite witness. Assay P2-E01 enumerates all 128 seven-coordinate access records and accepts the 14 satisfying the implication spine. Assay P2-E02 does the same for the 128 operational profiles separating rule soundness, execution, reachability and firing, accepting 12. The counts are implementation receipts, not evidence for the law.

Definition 5.2 (Exposure and rigidity calculus, FORMAL SCHEMA, VII-C022). Determination, exposure, recoverability, and admissibility are independent interface coordinates. *Exposure overread* occurs when a distinction visible at the interface is treated as determined by the substrate; *spurious rigidity* occurs when the absence of an admissible coarse-graining is read as a property of the object rather than of the declared interface.

Provenance. RECOVERED SBT (exposure/no-overread and information-loss recoverability). SHARPENED (the two named failure modes (exposure overread, spurious rigidity) as typed certificate conditions). *Canonical:* Foundations IV F11 Adequacy/No-Overread (Thm. 3.3), Def. 3.5 (hiddenness exposure), F34 Information Loss Normal Form (Thm. 8.3) [33]. *Relation:* Weaker and differently scoped: F11 and F34 are theorems about quotients; VII states a typed schema over interface coordinates and proves only the record-level separations. *Lean:* FoundationsVII.DomainState.AuxiliaryRecoveryCertificate.SpuriousRigidity, FoundationsVII.DomainState.same_determination_can_have_different_exposure; receipt axioms: none.

Converse and partition. Exposure is not determination, and recoverability is not permission. The two failure modes named in Definition 5.2 are the practically common ones, and both are errors about the interface mistaken for errors about the world.

Layer-specific bridge obligations. Spurious rigidity is the one most likely to be inherited incorrectly. An object with no admissible coarse-graining *under a declared interface* may have several under a different one, so a rigidity result transported between interfaces without a bridge naming both is unsound. Foundations IV’s access-quotient vocabulary is inherited here, and the obligation is to record which interface a quotient is taken over.

A worked reading of the record. Consider a package in which a content is expressible, present, and admissible, but not exposed, not recoverable, not reachable, and not occurrent. The record is coherent: every implication in the spine is satisfied, since the antecedents of the ones that would force more are false. What it describes is a permitted content that is really there, is not visible at the interface, has no route to it, and has not happened.

Four common misreadings of that record are each blocked by a different coordinate. It is not “latent” in a way that makes it available on demand, because *reachable* is false. It is not hidden in a way that could be uncovered by looking harder, because *exposed* is false and exposure is an interface property rather than an epistemic one. It is not merely unobserved, because *occurrent* is false and occurrence is a fact about the system rather than about an observer. And it is not inadmissible,

because `admissible` is true — permission and availability are different coordinates, and this record has one without the other.

That a single seven-bit record distinguishes four readings which ordinary usage runs together is the argument for Definition 3.1.

5.2 Admission as a typed transition system

Definition 5.3 (Lawful admission transitions, FORMAL SCHEMA, VII-C002). Admission, expiry, revocation, retraction, and rollback are typed transitions. Each requires source, budget, guard, and audit witnesses, and each has a typed effect on the admissibility coordinate.

Provenance. RECOVERED SBT (admissibility as a typed, witnessed claim). SHARPENED (the five-transition system with typed effects and append-only survival of history). *Canonical:* Foundations II Def. 3.1 (admissible claim) [31]; Foundations V E5 Reclosure Collapse (Thm. 5.5), repair-generator reachability record [34]; Lean adapters FT01, FT02, FT17. *Relation:* Differently scoped: Foundations II types single claims; VII types the transitions that change admissibility and proves their typed effects. *Lean:* `FoundationsVII.AdmissionOperation`, `FoundationsVII.TypedAdmissionStep.Lawful`, `FoundationsVII.TypedAdmissionStep.EffectTyped`, `FoundationsVII.TypedAdmissionStep.admit_sets_admissible`; receipt axioms: none.

Positive normal form. Admission is not a single act but a five-transition system, and the four non-admitting transitions are as typed as the admitting one. A framework with only “admit” cannot distinguish a claim that was withdrawn from one that was never made.

Converse and partition. A well-formed rule need not be executable. Well-formedness is a property of the record; executability additionally requires that the witnesses exist.

Exact finite witness. Assay P5-E01 enumerates the admission family over its declared carrier and confirms that each transition’s typed effect is determined by its witnesses.

5.3 Bootstrap

Theorem 5.4 (Bootstrap obstruction, CONDITIONAL THEOREM, VII-C003). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then a closed admission regime with no admitted seed and no reachable generator has no lawful first extension.*

Provenance. RECOVERED SBT (no reachable repair generator excludes endogenous first moves; no free distinction). SHARPENED (the obstruction stated over a declared closed regime, with its three escapes proved: two authorisations inside the closed regime (admitted seed, reachable generator) and one scope escape (an open family with external provision)). *Canonical:* Foundations V E1 Internalization (Thm. 5.1), `SixBirdsFoundationsV.NoReachableRepairGenerator` [34]; Foundations IV F12 No-Free-Distinction (Thm. 3.4), `SixBirdsMetaMath.FoundationsIV.Access.NoFreeDistinction.financed_refinement_retains_access` [33]. *Relation:* Differently scoped: E1 and F12 are stated inside one package; VII restates the obstruction over an admission regime and proves the two in-regime authorisations and the open-family escape. *Lean:* `FoundationsVII.AdmissionRegime.no_first_extension_without_seed_or_reachable_generator`, `FoundationsVII.NoGo.NGVII_01_no_first_extension_without_seed_or_generator`; receipt axioms: none.

Proof of Theorem 5.4. A lawful first extension requires the transition to be permitted and its authorisation to be witnessed. Authorisation is discharged in exactly three ways: an admitted seed, a reachable generator, or open external provision. The first two are excluded by hypothesis. The third requires the family to be open, and is excluded by closure. Since authorisation admits no fourth route in the declared regime, no lawful first extension exists. Removing any single hypothesis restores the corresponding route, which is what makes each escape an escape rather than a caveat. $\square$

Strictness. The obstruction is exact and its hypotheses are all three load-bearing: closure of the regime, absence of a seed, and absence of a reachable generator. Drop any one and the conclusion fails, which is what the escape routes below record.

Converse and partition. External provision is *not* forbidden. The theorem is scoped to a *declared closed* family, and three escapes are named: an admitted seed, a reachable generator, and open external provision. A reader who quotes the obstruction without the escapes has misstated it as a claim about what cannot happen, when it is a claim about what cannot happen *within a family that has declared itself closed*.

Exact finite witness. Assay P5-E06 enumerates the bootstrap family over its declared carrier, including all three escape cases, so the boundary is exhibited.

5.4 Seeds and prospective commitment

Definition 5.5 (Neutral provisioning certificate, FORMAL SCHEMA, VII-C004). A seed receives neutral-provisioning status only when its source, symmetry, task-blindness, preregistration, and control coordinates are recorded and the source is not system-generated.

Provenance. RECOVERED SBT (pre-outcome (prospective) evidence and shared-budget allocation records). SHARPENED (a typed neutral-provisioning certificate with two anti-laundering theorems). *Canonical:* Foundations V E10.1 Intention (Cor. 6.7: positive evidence restricts action support *before* the outcome), E15 Offline Reclosure (Thm. 8.2) shared-budget record [34]; Lean adapter FT04. *Relation:* Differently scoped: Foundations V prices pre-outcome evidence inside a cognitive package; VII types the seed a regime may admit. *Lean:* FoundationsVII.NeutralProvisioningCertificate, FoundationsVII.NeutralProvisioningAt, FoundationsVII.NeutralProvisioningCertificate.retrospective_stocking_cannot_earn_neutral_credit; receipt axioms: none.

Converse and partition. Neutrality here is *audit-relative*, not metaphysical independence. The certificate records that the seed was not tailored to the result it is about to support; it does not assert that the seed came from nowhere. Two anti-laundering results attach: a retrospectively stocked seed cannot earn neutral credit, and routing a system-generated source through an external label does not launder it.

Exact finite witness. Assay P2-E04 enumerates 256 prospective-credit assignments over pre-registration, precedence and control coordinates, accepting 129. CM-20, post-hoc prediction self-certification, is the named witness among the rejected assignments.

Definition 5.6 (Prospective commitment, FORMAL SCHEMA, VII-C006). A prospective commitment is a timestamped, preregistered, finite-budget rule that fixes later admission conditions *before* the target evidence is available.

Provenance. RECOVERED SBT (prospective (pre-outcome) credit). SHARPENED (a timestamped, pre-registered, finite-budget commitment record with the precedence certificate made explicit). *Canonical:* Foundations V E10.1 Intention (Cor. 6.7) [34]; Foundations I T-AOT-02 protocol trap (Thm. 5): no sustained asymmetry without a genuine drive; the temporal-precedence discipline VII adopts [30]. *Relation:* Differently scoped: VII adds the explicit event-precedence relation and refuses retrospective certification (NGVII-03) without assuming a global clock. *Lean:* `FoundationsVII.ProspectiveCommitment`, `FoundationsVII.CommitmentUse.ProspectivelyCertified`, `FoundationsVII.CommitmentUse.retrospective_predicate_cannot_discharge_prospective_certificate`; receipt axioms: none.

The prospective/retrospective distinction also motivates preregistration in scientific methodology: an analysis plan fixed before outcomes are inspected has a different evidential status from one reconstructed afterwards [22]. VII borrows that temporal discipline, but its finite-budget commitment record is its own formal object.

Converse and partition. Commitment does not guarantee reachability or success. It fixes the standard in advance; it does not promise the standard will be met. A commitment registered after the evidence is not a commitment, and the timestamp is what makes that checkable.

5.5 Protocol discipline

Three further entries govern how claims about these objects may be made at all. They are grouped here because they constrain the whole catalog rather than any one law: every result in Sections 6 to 8 is stated under them, and a result that violated one would be inadmissible regardless of whether its own hypotheses held.

The three answer three different questions. Definition 5.7 asks what licenses moving a result across a boundary. Definition 5.8 asks what a statement is claiming and against what standard it is judged. Theorem 5.9 asks how a law is tested at all — what would count as evidence for it, and what would count against.

Definition 5.7 (Bridge and semantic-withdrawal discipline, FORMAL SCHEMA, VII-C020). Every imported law or interaction claim carries: source and target maps; the hypotheses preserved, added, and lost; an audit path; its nonclaims; and *append-only* failure or withdrawal records. Citations alone never license transport.

Provenance. RECOVERED SBT (bridge records with source/target maps and preserved/lost hypotheses; exact and defective bridges). SHARPENED (append-only failure and withdrawal records; citation alone never licenses transport). *Canonical:* Foundations III §3–4 promotion and visibility bridges, §11 bridge policy [32]; lawful transfer, exact and defective bridge records [37] (Defs. 8.1–8.3, §8); Foundations IV F19 Object Persistence (Thm. 7.3) [33]. *Relation:* Differently scoped: the corpus bridges carry results between layers of one setting; VII’s discipline governs transport of *laws* between theories and adds withdrawal records. *Lean:* `FoundationsVII.BridgeContract`, `FoundationsVII.TransportAuthorization.licensed`, `FoundationsVII.TransportAuthorization.citation_only_is_not_licensed`; receipt axioms: none.

Strictness. Two clauses do the work. The records are append-only, so a bridge that failed or was withdrawn remains visible instead of vanishing — a later reader can see that a transport was attempted and refused. And a citation is not a licence: recording where something was said is different from recording that it may be used here, and the framework treats an uncertified bridge as unlicensed transport however well attested the source.

Definition 5.8 (Claim-grade and normative-specification discipline, FORMAL SCHEMA, VII-C024). Statements preserve their theorem, schema, calibration, conjecture, and philosophy grades; label

necessary and sufficient conditions explicitly rather than leaving the direction to context; and are judged against *normative specifications* rather than against illustrative examples.

Provenance. RECOVERED SBT (claim grades and explicit nonclaim records). SHARPENED (a typed grade-faithfulness record that refuses upgrade by finite evidence). *Canonical:* Foundations II Def. 3.1 (admissible claim), nonclaim records; Lean `SixBirds.AdmissibleClaim`, `SixBirds.NonclaimRecord` [31]. *Relation:* Identical in intent; VII adds the theorem that Lean or external finite evidence does not upgrade a claim to theorem or corollary grade. *Lean:* `FoundationsVII.ScienceAssetRecord.GradeFaithful`, `FoundationsVII.ScienceAssetRecord.lean_finite_evidence_does_not_upgrade_to_theorem`; receipt axioms: none.

Strictness. The last clause is the least obvious and the most consequential. An illustrative example shows that something can happen; a normative specification says what must be recorded for a claim to count. Judging a claim against examples permits an accumulation of suggestive cases to stand in for a specification that was never written.

Theorem 5.9 (Two-Theory World detector contract, LEAN DECIDABLE FINITE, VII-C025). *In the finite reference world, a detector contract records, for the interaction laws, append-only evidence together with non-empty signal, null and falsifier case lists and non-empty same-source, no-contact, scheduling and relabelling control lists, with a false-positive cost strictly above the false-negative cost. Over the declared carrier of 24 frozen scenarios and 27 named countermodels the contract is well formed and complete, and every fixture’s expected status is decided by the finite evaluator; all of this is decidable and is checked by evaluation.*

Provenance. RECOVERED SBT (signal/null/falsifier controls and calibrated null regimes). SHARPENED (one typed detector contract over the frozen reference world, decidable over its 24 + 27 fixtures). *Canonical:* Foundations III §9 countermodel atlas [32]; null regimes and trap checks, §4 [27]. *Relation:* Differently scoped: the corpus supplies the control discipline; VII binds it to a fixed finite reference world and makes the contract Lean-decidable. *Lean:* `FoundationsVII.DetectorContract`, `FoundationsVII.Models.Finite.phase1_detector_contract_wellFormed`, `FoundationsVII.Models.Finite.all_scenarios_pass`; receipt axioms: none.

Strictness. The four controls are what make the detector a detector rather than a description. Same-source, no-contact, scheduling and relabelling are precisely the four cheap ways to produce an apparent interaction, and a law that fires on any of them is mis-specified. The asymmetric cost is deliberate: a false positive — claiming an interaction that is not there — is treated as the more expensive error.

Converse and partition. Decidability is a property of the declared carrier, not of interaction in general. The contract decides the fixtures it was frozen against; it is not a decision procedure for whether two arbitrary theories interact.

Exact finite witness. The contract’s own carrier is the 24 frozen scenarios and 27 named countermodels of Section 10. Every fixture carries its expected status and its recorded hash, and both implementations agree on all 51.

Layer-specific bridge obligations. Cluster A inherits from Foundations II’s admissibility vocabulary, and the inheritance is not free. Three obligations attach. A VII admission transition must be shown to specialise, not merely resemble, the Foundations II notion — the specialisation is recorded per declaration rather than argued once. The seven access coordinates of Definition 3.1

include three that specialise inherited status machinery (`exposed`, `recoverable`, `admissible`) and four that do not; only the first three carry inherited results, and the other four must be established here. And the bootstrap obstruction is stated over VII’s own regime, so no Foundations II result about admissibility transfers to it without a recorded bridge.

Formal correspondence and boundary. This cluster corresponds to the `FoundationsVII.Access` and `FoundationsVII.Core.Admission` module families, together with `FoundationsVII.Core.Protocols` for Section 5.5. All declarations are kernel-verified in the sense of Section 2.3: they elaborate, and every theorem among them carries an executed `#print axioms` receipt. Per Remark 2.1 this leaves every `FORMAL SCHEMA` in this cluster a schema.

6 Cluster B — Contact and join

This is the largest cluster, and the one the title names. Ten catalog entries govern when two theories may touch, when touching produces a joint object, what that object must retain of its parents, and what it costs. The cluster is ordered so that the negative results arrive early: a reader who forms the lattice intuition of Remark 4.1 and only later meets its refutation has to unlearn, and unlearning is expensive.

6.1 Contact without join

Definition 6.1 (Join-entry record, `FORMAL SCHEMA`, VII-C007). Entry to the join calculus requires: both parent packages; a typed contact surface; witnessed contact or an explicit pending status; source and licensing ledgers; a budget; and an audit. A join-entry record is *well formed* when every field is present and correctly typed.

Provenance. `RECOVERED SBT` (instrument-indexed claim records and typed contact witnesses). `SHARP-ENED` (a join-entry normal form separating well-formedness from existence). *Canonical:* Foundations III §11 instrument-indexed claim records; Lean adapter `FT06 SixBirdsIII.InstrumentClaimRecord` [32]; Foundations V D1 repair join (Def. 3.1), contrasted, not reused [34]. *Relation:* Differently scoped: Foundations III records claims under one instrument; VII records the entry conditions for a join of two packages. *Lean:* `FoundationsVII.JoinEntryRecord`, `FoundationsVII.JoinEntryRecord.WellFormed`, `FoundationsVII.JoinEntryRecord.wellFormed_has_two_parents`; receipt axioms: none.

Positive normal form. Definition 6.1 is the admission form for the whole cluster. Every later result presupposes a well-formed entry record, and no result is stated for an interaction that has not been recorded.

Converse and partition. Well-formedness of the record does *not* prove that a join exists. The record establishes that the question is well posed, not that the answer is yes. This is the first place the lattice intuition misleads: for the repair join of Foundations V, posing the question and answering it are the same act, because the operation is total on its lattice. Here they come apart.

Theorem 6.2 (Peer transport without join, `CONDITIONAL THEOREM`, VII-C016). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then certified content may become available to a peer through a typed contact and transport record without a joint theory being formed, provided destination ownership and provenance are preserved.*

Provenance. RECOVERED SBT (repair transport with transport tokens; instrument visibility). SHARPENED (five contact modes with licensing evidence and transport-without-join theorems). *Canonical:* Foundations V E13 Repair Transport (Thm. 7.4), SixBirdsFoundationsV.TransportTokenRecord [34]; Foundations III §11 visibility bridges [32]. *Relation:* Differently scoped: E13 transports repairs inside a cognitive carrier; VII proves that certified transport to a peer never by itself forms a composite. *Lean:* FoundationsVII.PeerTransport.WithoutJoin, FoundationsVII.PeerTransport.withoutJoin_is_certified_transport, FoundationsVII.peer_contact_can_transport_without_join; receipt axioms: none.

Proof of Theorem 6.2. Availability of certified content at the destination is discharged by the transport record together with the licensing ledger; neither mentions a composite package. Formation of a join additionally requires composite objecthood, which no clause of the transport record supplies. Ownership and provenance preservation are exactly the conditions under which the transported content remains attributable to its source, so the destination acquires content without acquiring authorship of it. The finite counterpart is existential rather than exclusionary: P3-E01 contains accepted profiles in which transport credit holds and no composite is formed, which is what establishes the separation. It also contains accepted profiles carrying both, since transport and composite formation are compatible — the theorem says transport does not *entail* a composite, not that the two exclude one another. $\square$

Strictness. This separates Definition 3.4 from Definition 3.5. Availability of content across a boundary is cheap; joint objecthood is not. Systems that exchange records, share carriers, or transport payloads are interacting, and none of that yet produces a third theory.

Converse and partition. Peer transport is *not* joint objecthood and not strict novelty. A reader tempted to read successful transport as evidence of a join should note that the countermodel battery separates the two directly: a shared carrier without lawful peer transport, and lawful transport without any composite object, are both exhibited.

Exact finite witness. Assay P3-E01 enumerates the contact and transport family exhaustively over its declared carrier of 256 profiles, accepting 54. Among the accepted are profiles with transport credit and no composite — the separating cases — and profiles with both, which is why the theorem is stated as a non-entailment rather than as an exclusion.

6.2 The status calculus

Definition 6.3 (Join status, FORMAL SCHEMA, VII-C008). Within a declared finite interface family, a join attempt receives exactly one status: no evidenced contact; evidenced contact; common refinement; lawful composite; strict join; typed obstruction; or certified non-interaction.

Provenance. SHARPENED (an exact seven-way status classification over a declared interface family). RECOVERED SBT (the status-family and coverage vocabulary). *Canonical:* Foundations III §5 status and defect semantics, Thm. 5 Status-Family Separation [32]; Foundations V E11 Institutional Rewrite (Thm. 7.1) obstruction status; Lean adapters FT09, FT10 [34]. *Relation:* Differently scoped: Foundations III classifies cell status inside one calculus; VII classifies join attempts between packages and proves the partition only over the declared family. *Lean:* FoundationsVII.JoinStatus, FoundationsVII.JoinAssessment.classify, FoundationsVII.Models.Finite.Phase3.status_partition_complete; receipt axioms: none.

Positive normal form. Over the declared family the statuses are mutually exclusive and jointly exhaustive. That is what gives “the join failed” content rather than mere absence: a failed attempt names the status it received.

Converse and partition. The taxonomy is *not* universal without a completeness theorem, and none is claimed. Outside the declared interface family the seven statuses are not asserted to be jointly exhaustive.

Exact finite witness. Assays P3-E02, P5-E03 and P5-E11 enumerate the status calculus over their declared carriers; every case within those carriers receives exactly one status.

6.3 Strict join and the anti-product witness

Theorem 6.4 (Strict join certificate, CONDITIONAL THEOREM, VII-C009). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then a strict join requires all three of: composite objecthood; recoverable parent retention; and a distinction or operation that factors neither through either parent alone nor through the declared product or common refinement of the two.*

Provenance. RECOVERED SBT (strict extension as non-factorization; strict self-extension). SHARPENED (three necessary conjuncts of a strict-join certificate, with factorization through the declared product and common refinement excluded). *Canonical:* Foundations IV F8 Strict Extension (Lemma 4.3), F51 Unification as Common Refinement (Thm. 8.18) [33]; Foundations III Thm. 12 Strict-Extension Nonfactorization [32]; Foundations V E1 SixBirdsFoundationsV.StrictSelfExtension [34]; strict-test admissibility on fixed support [40]. *Relation:* Differently scoped: the corpus lemmas characterise strictness of one extension; VII’s certificate requires composite objecthood, parent retention, source independence and a settled budget in addition, and its anti-product witness excludes factorization through the declared product and common refinement; only necessity of the three named conjuncts is proved. *Lean:* FoundationsVII.StrictJoinEvidence.Certified, FoundationsVII.StrictJoinEvidence.certified_has_anti_product_novelty, FoundationsVII.AntiProductWitness.valid_is_nonfactorizing; receipt axioms: none.

Proof of Theorem 6.4. Necessity of each conjunct is shown by exhibiting a configuration satisfying the other two and failing strictness. Without composite objecthood there is no object to be strict about, and the configuration is a contact record. Without recoverable parent retention the candidate is not a join of *those* parents, since the parents are not recoverable from it. Without the non-factoring witness every operation of the candidate factors through a parent, the product, or the common refinement, so the candidate’s content is already determined by the baseline and adds nothing. Two of the three omission configurations are named witnesses in the formal development — anti-product novelty without objecthood, and anti-product novelty without parent retention; the third — objecthood and retention with no anti-product witness — is refused by NGVII-07 for a mere product or common refinement and by the relabelling, scheduling and coarsening refusals of the anti-product witness. P3-E03 enumerates the omission attacks over its declared carrier of 65,536 canonical cases, of which 32,770 satisfy the law. $\square$

Strictness. The third conjunct is load-bearing, and it is where the categorical reading dies. Excluding factorization through the parents alone is not enough: an operation could still factor through their product or common refinement, and a composite whose every operation did so would carry no content beyond what the two parents already jointly determine. Both exclusions are therefore required, and the witness must be exhibited rather than argued.

Converse and partition. Two converses fail. Strict join does *not* imply a directionality arrow — strictness and direction are separate certificates, and Section 8 keeps them apart. And parent strictness alone does *not* prove joint strictness: two theories each of which is a strict extension of something need not join strictly.

Exact finite witness. Five assays bear on this entry (P3-E03, P3-E10, P5-E01, P5-E02, P5-E07). The strictness carrier alone contains 65,536 canonical cases, of which 32,770 satisfy the declared law. Within that carrier a directionless strict case exists — strictness without a drive certificate — which is the finite counterpart of the first converse above.

6.4 Retention, and a refuted monotonicity

Remark 6.5 (Parent refinement and join descent, REFUTED CANDIDATE, VII-C030). The unconditional claim that refining a parent preserves join descent is **refuted**. What survives is a four-way typed classification: parent refinement may *preserve*, *strengthen*, *weaken*, or *destroy* the join certificate, and which occurs is determined by the retention record rather than by the refinement alone.

Provenance. RECOVERED SBT (object persistence under extension). SHARPENED (the unconditional monotonicity claim refuted by a minimized falsifier; the four-way typed classification). *Canonical:* Foundations IV F19 Object Persistence (Thm. 7.3), `SixBirdsMetaMath.FoundationsIV.StatusRecordsCoherence.ObjectPersistence.persistence_extension_retains_old` [33]; Foundations IV F2 Descent-Repair Normal Form (Thm. 5.2) [33]. *Relation:* Differently scoped: F19 states what persists under a lawful extension; VII shows that refining a parent may destroy an admissible join (CM-22) and types the four outcomes. *Lean:* `FoundationsVII.RefinementEffect`, `FoundationsVII.no_unconditional_join_monotonicity_under_refinement`, `FoundationsVII.refinement_effects_have_all_four_controls`; receipt axioms: none.

Converse and partition. The refutation is not a gap awaiting repair. A valid descent certificate does not rescue a refinement that destroys compatibility or strictness; the destroy case is real and is exhibited: CM-22 (scenario TTW-S24, expected status `REFINEMENT_DESTROY_JOIN`) and CM-05 (scenario TTW-S09, expected status `RETENTION_OBSTRUCTION`); assay P3-E07 (576 raw, 138 accepted) populates all four classes, and P5-E05 (2,048 raw, 496 accepted) composes them with descent fidelity. Retention alone is likewise not strict novelty — a composite may retain everything of both parents and add nothing.

Exact finite witness. Assays P3-E07 and P5-E05 enumerate the refinement and retention family over their declared carriers and populate all four classes, which is what makes the classification a partition rather than a hedge.

6.5 Source independence

Definition 6.6 (Source-independence gate, FORMAL SCHEMA, VII-C010). An independence-sensitive join claim requires an ancestry witness *separate* from behavioral non-factorization. Native and bridged records retain their distinct source identities.

Provenance. RECOVERED SBT (carried source records and source-of-truth provenance). SHARPENED (the gate separating ancestry independence from behavioural non-factorization). *Canonical:* Foundations V Def. 3.3 carried record, D3 carried-source exclusion (Prop. 3.9), `SixBirdsFoundationsV.CarriedSource` [34]; Foundations III source-of-truth records, §3 [32]. *Relation:* Differently scoped: the corpus certifies

that a record is carried; VII requires an ancestry witness separate from behaviour before independence-sensitive credit. *Lean*: `FoundationsVII.SourceIndependenceGate.IndependenceSensitiveCredit`, `FoundationsVII.SourceIndependenceGate.behavioral_nonfactorization_alone_is_insufficient`; receipt axioms: none.

Strictness. Behavioural non-factorization and source independence are different properties, and the gate refuses to infer either from the other. Two theories may be behaviorally independent while sharing an ancestor, and may share no ancestor while factoring through a common construction.

Converse and partition. Source independence is *not* sufficient for strict novelty, nor for causal independence.

Neighboring formalisms make different parts of this separation precise: provenance calculi track derivational lineage, while causal graphical models distinguish shared causes and conditional independences [13, 23]. Neither notion is silently substituted for the ancestry witness required here.

Theorem 6.7 (Common-origin non-transfer, CONDITIONAL THEOREM, VII-C005). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then common origin, common carrier and common instrument, even jointly, do not entail shared access; shared access does not entail source independence; and a shared lineage defeats, rather than supplies, independence-sensitive join credit.*

Provenance. RECOVERED SBT (common-source explanation and descent-to-common-source obstruction). SHARPENED (three proved clauses, each carried by a named witness or refusal (CM-01, CM-26)). *Canonical*: Foundations IV F49 Common-Source / Nonlocal Correlation Normal Form (Thm. 8.16), F17 Objectivity as Common Quotient (Thm. 7.1) [33]. *Relation*: Differently scoped: F49 characterises when a common source explains a correlation; VII refutes the inference from shared origin, carrier and instrument to shared access, from shared access to source independence, and from a shared lineage to independence-sensitive credit. *Lean*: `FoundationsVII.PackageAccess.common_origin_countermodel`, `FoundationsVII.PackageAccess.shared_access_does_not_imply_source_independence`, `FoundationsVII.SourceIndependenceGate.same_lineage_fails_independence_sensitive_credit`; receipt axioms: none.

Proof of Theorem 6.7. Each clause is carried by a recorded witness or a proved refusal. For the first, a single pair of packages sharing source, carrier and instrument whose access records nevertheless differ refutes the entailment to shared access in all three directions at once (CM-01). For the second, a pair with shared access and a common source is not source-independent, since independence is a difference of source. For the third, the source-independence gate refuses independence-sensitive credit to any pair of the same lineage, so a behavioural difference produced inside one lineage cannot earn it (CM-26). The related refusal that a total-lens result does not transfer to a partial self-owned target without a certified adapter (CM-12) is NGVII-04, stated in Section 16. $\square$

Strictness. This negative result blocks an inference that shared provenance invites: provenance is visible, and visibility tempts the step to interaction. The theorem blocks the inference from shared provenance to access, from access to independence, and from a shared lineage to independence-sensitive credit.

Converse and partition. Common origin may still be *relevant evidence* when correctly typed; the theorem forbids the entailment, not the evidence. Countermodels CM-01 and CM-26 exhibit the separations; CM-12 belongs to NGVII-04.

6.6 Budget: joins are paid for

Definition 6.8 (Join budget ledger, FORMAL SCHEMA, VII-C011). A join ledger prices interface access, parent retention, joint novelty, observer and instrument occupancy, cross-costs, and explicit refunds. A claim of zero total cost is admissible only when certified.

Provenance. RECOVERED SBT (priced access, binding exposure budgets, positive move costs; residual currency). SHARPENED (a typed join ledger with five cost kinds, refunds and certified zero-cost channels). *Canonical:* Foundations V E6 Attention (Thm. 6.1), E9 Curiosity (Thm. 6.5), `SixBirdsFoundationsV.BudgetFeasible`, `SixBirdsFoundationsV.BindingExposureBudget` [34]; adequacy residual and blind-spot currency (Def. 4.2) [26]. *Relation:* Differently scoped: Foundations V prices access moves inside one package; VII prices the formation of a join and forbids uncertified zero cost. *Lean:* `FoundationsVII.JoinCostKind`, `FoundationsVII.JoinCostEntry.Credited`, `FoundationsVII.JoinPaymentLedger.FullyPaid`, `FoundationsVII.JoinCostEntry.positive_cost_without_payment_or_zero_channel_is_not_credited`; receipt axioms: none.

Positive normal form. Pricing is not bookkeeping added for tidiness. Clause (iii) of Definition 3.5 is this ledger, and without it the framework admits joins that pay for nothing — the failure the next result forbids.

Converse and partition. No universal conserved interaction currency is asserted. The ledger is a local accounting discipline over declared resources, not a conservation law, and Section 8 refutes the universal version explicitly.

Nor is the ledger, without an application bridge, a thermodynamic cost statement. Landauer’s analysis and Bennett’s review concern physical information processing [16, 4]; they become relevant only for an instantiation that maps the ledger’s declared resources to physical operations and discharges the corresponding hypotheses.

Theorem 6.9 (Observer and instrument occupancy, CONDITIONAL THEOREM, VII-C029). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then an observer or instrument that uses bounded shared capacity cannot certify native or endogenous formation while its occupancy is under-charged — charged occupancy below actual occupancy — or while its source role or cost is left off the ledger.*

Provenance. RECOVERED SBT (no-overreading of suppressed content; priced observer exposure). SHARPENED (the unpriced-occupancy refusal with the external-observer and zero-occupancy escapes proved). *Canonical:* Foundations III Thm. 23 No-Overreading Suppression [32]; Foundations V E6/E9 priced access, `SixBirdsFoundationsV.PositiveAccessMoveCosts` [34]. *Relation:* Differently scoped: Thm. 23 forbids claims resting on unbridged suppressed content; VII forbids native or endogenous credit while the observer’s own occupancy is under-charged. *Lean:* `FoundationsVII.ObserverOccupancyRecord.NativeFormationCredit`, `FoundationsVII.ObserverOccupancyRecord.unpriced_observer_invalidates_native_formation_credit`; receipt axioms: none.

Converse and partition. Observation is not prohibited; *unrecorded load-bearing* observation is. An instrument that consumes declared capacity and is recorded as doing so raises no obstruction. What the theorem forbids is the certificate that omits it.

6.7 No free join

Theorem 6.10 (No-free-join family, CONDITIONAL THEOREM, VII-C035). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then a join cannot: create the first capability required to execute itself; manufacture strict joint information by relabelling; or receive positive-cost joint formation for nothing.*

Provenance. RECOVERED SBT (no free distinction; financed refinement). SHARPENED (three cheap routes to a fake join closed by three different gates, with the escapes named). *Canonical:* Foundations IV F12 No-Free-Distinction (Thm. 3.4), `SixBirdsMetaMath.FoundationsIV.Access.NoFreeDistinction.finance_refinement_retains_access` [33]; Foundations V E1 Internalization (Thm. 5.1) [34]. *Relation:* Differently scoped: F12 forbids a functorial gate from producing a strict distinction for free; VII forbids a join from bootstrapping, relabelling, or forming unpaid. *Lean:* `FoundationsVII.NoGo.NGVII_09_no_positive_cost_join_credit_without_payment`, `FoundationsVII.AntiProductWitness.relabel_only_fails`, `FoundationsVII.NoGo.SelfBootstrapJoinProfile.Lawful`; receipt axioms: none.

Proof of Theorem 6.10. The three clauses rest on different gates, and each follows from a gate already established. The first is Theorem 5.4 applied to the join’s own execution: a capability required to execute the join cannot be created by that execution, since the authorisation would have to precede itself in the recorded precedence. The second follows from Theorem 6.4’s third conjunct: a relabelling factors through its parent by construction, so it supplies no non-factoring witness. The third is Definition 6.8 clause (iii): positive-cost formation with no discharge leaves the ledger unbalanced, and an unbalanced ledger is not a certificate. In the formal development the three clauses are carried by three separate records — the self-bootstrap profile and its named control, the anti-product witness, and the payment credit — none of which is derived from another; that the three rest on different gates is the observation above, not a formalised independence theorem, and the general form of the first clause is the prose argument from Theorem 5.4. $\square$

Strictness. The three clauses close the three cheap routes to a fake join — bootstrap, relabelling, and unpaid formation — and, as the proof records, they rest on three different gates.

Converse and partition. The family is scoped, and its escapes are named rather than implied: it permits admitted seeds, declared bridges, and *proven* zero-cost channels. A zero-cost channel is not forbidden; an *uncertified* claim of zero cost is. Stating the obstruction without these escapes would present it as a stronger result than it is.

Exact finite witness. Assays P3-E05 and P5-E06 enumerate the no-free-join family over their declared carriers, including the escape cases, so that the boundary between the obstruction and its permitted channels is exhibited rather than asserted.

Layer-specific bridge obligations. This cluster carries the heaviest inheritance load in the paper, and the obligations are correspondingly explicit. The word *join* is inherited from Foundations V in a different sense and must be kept apart at every use (Remark 4.1); the budget ledger of Definition 6.8 specialises the inherited currency vocabulary without inheriting any conservation claim, and the refutation of a universal conserved contact degree (Remark 8.6) is what makes that separation load-bearing rather than cautious; and the anti-product witness of Theorem 6.4 is stated against a *declared* comparison baseline, so no categorical result transfers to it from outside without a bridge naming that baseline.

Formal correspondence and boundary. The entries of this cluster correspond to the `FoundationsVII.Join` and `FoundationsVII.Contact` module families, together with the no-go modules carrying the escape theorems. All are kernel-verified in the sense of Section 2.3: the declarations elaborate, and every theorem among them carries an executed `#print axioms` receipt. Per Remark 2.1, this does not raise any FORMAL SCHEMA in this cluster to a theorem.

7 Cluster C — Enablement and descent

Nine entries govern what it means for one theory to make another possible, and how that differs from causing it, containing it, or determining it. This is where the paper’s most easily-abused notion lives: enablement is the relation reached for when one wants to say that a lower layer explains a higher one, and the cluster’s work is to type it precisely enough that the inference cannot be made for free.

7.1 Enablement and its attribution

Definition 7.1 (Enablement record, FORMAL SCHEMA, VII-C012). An enablement record identifies the operation made available, its source, the execution witness, the budget, the target effect, and the attribution — to theorist, carrier, peer, observer, environment, endogenous system, or a recorded mixed source.

Provenance. RECOVERED SBT (enablement (agenthood) distinguished from within-layer causation (agency)). SHARPENED (enablement promoted to a typed record with a seven-valued attribution field and a hidden-executor refusal). *Canonical:* §1.3 and §11.2 (causation versus enablement) [27]; Foundations V E10 interaction status (imports the agenthood distinction) [34]. *Relation:* Stronger: the corpus draws the distinction in prose and operationalises it in a Markov setting; VII types it and proves that a hidden executor defeats attribution. *Lean:* `FoundationsVII.EnablementRecord`, `FoundationsVII.EnablementSource`, `FoundationsVII.AttributedEnablement.WellFormed`, `FoundationsVII.AttributedEnablement.hidden_execution_defeats_attribution`; receipt axioms: none.

Constraint-closure accounts of biological organization describe constraints as acting on processes while mutually depending on and contributing to one another’s maintenance [19]. That literature is a substantive comparison class, but VII’s typed enablement relation is not claimed to formalize biological closure.

Positive normal form. Attribution is a field of the record, not an interpretation laid over it. The framework therefore distinguishes “this layer was enabled” from “this layer can be described as enabled”, and refuses the second where the first is not recorded.

Converse and partition. Enablement does not imply descent, and does not imply sufficient determination. These are the two inferences the term invites, and both are blocked. A separate result exhibits enablement without factorization or descent, so the separation is witnessed rather than asserted.

Exact finite witness. Assays P4-E01 and P5-E11 enumerate the attribution family over their declared carriers.

Theorem 7.2 (Enablement without descent, CONSTRUCTIVE COUNTERMODEL, VII-C034). *There exist enablement separation profiles in which enablement holds and is recorded as load-bearing while no descent factorization of the enabled object through the enabler is recorded. Enablement may also*

be necessary but insufficient: *a profile exhibits enablement recorded as necessary and not sufficient, with alternative determinants present.*

Provenance. RECOVERED SBT (enablement is not causation; descent as a separate certificate). SHARPENED (constructive Lean witnesses of enablement without descent and of necessary-but-insufficient enablement; the minimized falsifiers CM-06 and CM-07 in the reference world). *Canonical:* §11.2 [27]; Foundations III Thm. 30 Descent-Square Recovery [32]. *Relation:* Identical in the distinction drawn; VII supplies the explicit witnesses the corpus argues for. *Lean:* FoundationsVII.EnablementSeparation.EnablementWithoutDescent, FoundationsVII.constructive_enablement_without_descent, FoundationsVII.constructive_necessary_but_insufficient_enablement; receipt axioms: none.

Strictness. Load-bearing is essential to the witness. An enabler that merely accompanies the enabled object separates the two notions trivially and proves nothing; the profile records the enablement as load-bearing, and shows that even then no descent factorization is recorded. The witnesses are Boolean separation profiles, not full enablement records; the typed record of Definition 7.1 is a separate object.

Converse and partition. The countermodel refutes *named* claims, not a mood (CM-06, CM-07; scenario TTW-S03, expected status PROSPECTIVE_ADMISSION; assays P4-E05, 512 raw, 128 accepted, and P5-E05, 2,048 raw, 496 accepted). It kills the entailment “ $\mathcal{T}_1$ enables $\mathcal{T}_2$, therefore $\mathcal{T}_2$ descends from $\mathcal{T}_1$ ”, and separately the inference from necessity to sufficiency. It says nothing about settings in which descent independently holds.

Theorem 7.3 (Endogenous enablement criterion, CONDITIONAL THEOREM, VII-C013). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then enablement is endogenous only when the system carries, reaches, executes, and audits the enabling generator within the declared boundary, pays for it, and no theorist or observer executor is hidden.*

Provenance. RECOVERED SBT (no reachable repair generator excludes the endogenous family; carried records). SHARPENED (an exact biconditional criterion with hidden-theorist and hidden-observer clauses). *Canonical:* Foundations V E1 Internalization (Thm. 5.1), SixBirdsFoundationsV.NoReachableRepairGeneratorExcludesEndogenousFamily, Def. 3.3 carried record [34]. *Relation:* Stronger: Foundations V gives the exclusion; VII gives the full criterion as an equivalence and proves the two hidden-executor controls are independent of the other conjuncts. *Lean:* FoundationsVII.EndogenousEnablementProfile.Eligible, FoundationsVII.EndogenousEnablementProfile.eligible_iff_exact_criterion; receipt axioms: none.

Proof of Theorem 7.3. Endogeny is the conjunction of the recorded conditions: the enabling generator is carried, reachable, executed, audited and budgeted within a closed declared boundary; no theorist executor and no observer executor is hidden; and any environmental input is accounted. Each is a field of the enablement record, so the criterion is decidable over any declared carrier in which those fields are populated, and P4-E02 decides it over 2,048 profiles. The two negative controls — theorist-triggered and observer-triggered — satisfy every condition except the hidden-executor clause each violates, which shows that clause is not implied by the others and therefore cannot be dropped. $\square$

Strictness. The hidden-executor clause does the work. A system credited with endogenous formation while a theorist quietly performs the enabling step has not formed anything endogenously; it has been assembled. The criterion is exact, and its finite counterpart is decidable over the declared carrier.

Converse and partition. Endogenous does not mean uncaused or environmentally isolated. Accounted environmental input is compatible with endogeny; what is incompatible is *unaccounted* input. A system may be thoroughly coupled to its environment and still form endogenously, provided the coupling is on the books.

Exact finite witness. Assays P4-E02 and P5-E09 enumerate the endogeny family, including the theorist-triggered and observer-triggered negative cases.

7.2 Birth and the contact surface

Definition 7.4 (Birth classification, FORMAL SCHEMA, VII-C014). Layer birth is classified by who performs the closure, what new object survives, whether participants are created or only related, and whether the birth is recorded at a contact surface.

Provenance. RECOVERED SBT (the primitive taxonomy of layer-birth classes by activation of packaging, drive and staging). SHARPENED (birth classified additionally by who performs the closure and whether participants are created or only related). *Canonical:* canonical layer-birth classes, §2 [28]. *Relation:* Differently scoped: the corpus classifies births by primitive activation and separates them observably; VII adds the closure-agency and participant coordinates and proves that contact alone can remain relation-only. *Lean:* FoundationsVII.BirthRecord, FoundationsVII.BirthRecord.WellFormed, FoundationsVII.contact_can_remain_relation_only, FoundationsVII.every_inherited_birth_class_activates_packaging; receipt axioms: none.

Converse and partition. Contact does not itself constitute birth. This carries the Cluster B separation into Cluster C: two theories meeting, even lawfully and productively, does not by itself bring a third into existence.

Exact finite witness. Assay P4-E03 enumerates 1,280 closure-agency and birth profiles and accepts 160.

7.3 Transmission and descent fidelity

Theorem 7.5 (Typed transmission, CONDITIONAL THEOREM, VII-C015). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then upward, downward, and peer transmission are typed transmission records with explicit payload, loss, ambiguity, and source fields, and a pure downward selection — one with no external insertion and with the provenance of lower-layer facts preserved — cannot create lower-layer facts that were not present before the selection.*

Provenance. RECOVERED SBT (descent through a quotient; descent-square recovery; downward influence). SHARPENED (typed transmission records in three directions and the no-creation law for pure downward selection). *Canonical:* Foundations III Thm. 30 Descent-Square Recovery, Thm. 6 Descent Through Quotient [32]; Foundations IV F2 Descent-Repair Normal Form (Thm. 5.2) [33]; Foundations I §10.3 downward influence [30]. *Relation:* Differently scoped: the corpus results are about one map descending through one quotient; VII types the transmission record and proves that a pure downward selection creates no lower-layer fact. *Lean:* FoundationsVII.TransmissionRecord, FoundationsVII.TransmissionRecord.PureDownwardSelection, FoundationsVII.TransmissionRecord.pure_downward_selection_cannot_create_absent_lower_fact; receipt axioms: none.

Proof of Theorem 7.5. A pure downward selection is a well-formed downward transmission record with no external insertion whose lower-fact provenance is preserved: any lower-layer fact present after the selection was either present before it or externally inserted. With external insertion excluded, a fact present after the selection was present before it; hence a fact absent before is absent after. The payload, loss and ambiguity fields are explicit bookkeeping fields of the record and play no role in this step. $\square$

Strictness. The final clause is the fidelity law. Selection at a higher layer redistributes attention over what is already there; it does not manufacture substrate. A framework without this law admits explanations in which the act of describing a system populates it.

Converse and partition. Construction direction is not automatically temporal directionality. That a construction proceeds downward says nothing about time’s arrow, and Section 8 keeps the two apart with a separate certificate.

Exact finite witness. Assays P4-E04 and P5-E05.

Theorem 7.6 (Enablement-chain composition, CONDITIONAL THEOREM, VII-C027). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then enablement relations compose only when source types, execution witnesses, budgets, and residual obligations all compose. Chain order may carry holonomy.*

Provenance. ELEMENTARY (associativity and additive accumulation of resource deltas). RECOVERED SBT (holonomy of chain order). SHARPENED (composition credit conditioned on source, execution, budget and residual compatibility). *Canonical:* Foundations IV F3 Holonomy-Memory Repair Normal Form (Thm. 5.4) [33]; §11.2 [27]. *Relation:* Differently scoped: F3 concerns route residue of one carrier; VII conditions the composition of two enablement links and exhibits order sensitivity. *Lean:* FoundationsVII.EnablementLink.Composable, FoundationsVII.CompositionCredit.Valid, FoundationsVII.EnablementLink.compose_accumulates_cost, FoundationsVII.enablement_chain_can_be_order_sensitive; receipt axioms: none.

Converse and partition. Enablement is *not* assumed transitive. Transitivity is a conclusion available under the four composition conditions, not a property of the relation, and the holonomy clause records that the order in which a chain is traversed can leave a residue.

Exact finite witness. Assay P4-E06 enumerates 41,472 two-link composition profiles over source, budget, residual and audit coordinates, of which 10,449 compose.

7.4 Reachability and the force of a negative

Theorem 7.7 (Reachability and horizon, CONDITIONAL THEOREM, VII-C021). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then lawfulness, non-vacuity, reachability, firing, and occurrence are separate, and finite-horizon non-occurrence licenses only horizon-qualified conclusions unless the search family is closed, the detector’s power is established, and the null is complete through and beyond the declared horizon.*

Provenance. RECOVERED SBT (reachability records; horizon-qualified negative force). SHARPENED (the lawful / non-vacuous / reachable / fired / occurrent spine and the closed-family negative certificate). *Canonical:* Foundations V E5 Reclosure Collapse (Thm. 5.5) reachability records [34]; the scope discipline of the no-go theorems [38]. *Relation:* Differently scoped: VII separates five operational statuses and proves both the horizon qualification and its escape. *Lean:* FoundationsVII.OperationalProfile.Coherent, FoundationsVII.OperationalProfile.no_occurrence_from_reachability_alone, FoundationsVII.HorizonObservation.ClosedFamilyNegativeCertificate; receipt axioms: none.

The vocabulary intersects classical safety/liveness and model-checking accounts of state-transition systems, including bounded model checking [2, 8, 7]. The five operational statuses and the coverage-qualified negative certificate are VII-specific; the external comparison reinforces rather than removes the requirement to state the searched carrier and horizon.

Converse and partition. Non-occurrence alone is not impossibility. Something that did not happen within a declared horizon may still be reachable, and the theorem forbids upgrading the observation into a modal claim.

Exact finite witness. Assay P5-E08 enumerates 4,096 reachability profiles under horizon, closure and detector qualification and accepts 288. CM-18, a finite negative search with an open family, is the named witness.

Definition 7.8 (Negative-result quantifier discipline, FORMAL SCHEMA, VII-C023). The force of a negative result is fixed by its quantifiers and by the closure of its search family. A point null, a bounded search, an exhaustively enumerated finite family, and an unrestricted universal claim have four different forces and are not interchangeable.

Provenance. ELEMENTARY (the four quantifier forces of a negative result). RECOVERED SBT (nonclaim records and scoped no-go statements). SHARPENED (a typed licensing record for negative evidence). *Canonical:* Foundations II nonclaim records [31]; scoped no-go statements with escape routes [38]. *Relation:* Identical in logical content; VII records the force as a typed grade with licensing theorems. *Lean:* FoundationsVII.NegativeEvidenceRecord, FoundationsVII.NegativeEvidenceGrade, FoundationsVII.NegativeEvidenceRecord.point_null_does_not_license_unrestricted; receipt axioms: propext.

Strictness. This discipline governs how every no-go in this paper may be quoted. A result established by enumerating a declared family has exactly the force of that family, and stating it without the family inflates it into a claim the evidence does not support.

Converse and partition. A failed instance never refutes an unrestricted existential claim. One search that came back empty is one search.

Theorem 7.9 (Coverage-qualified non-interaction, CONDITIONAL THEOREM, VII-C033). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then certified non-interaction is a positive, family-scoped result only when contact channels, detector power, budget, horizon, and search-family closure are all declared.*

Provenance. RECOVERED SBT (coverage-qualified obstruction status). SHARPENED (certified non-interaction as a positive, family-scoped certificate with an explicit outside-family escape). *Canonical:* Foundations V E11 Institutional Rewrite (Thm. 7.1), SixBirdsFoundationsV.E11_NCTDObstruction [34]. *Relation:* Differently scoped: E11 states an obstruction inside institutional rewrite; VII types the certificate that licenses a scoped null between theories. *Lean:* FoundationsVII.ContactNullProfile.CertifiedNonInteraction,

FoundationsVII.ContactCoverageCertificate.Certified, FoundationsVII.DeclaredContactCase.e
xhaustive_declared_family_excludes_contact; receipt axioms: none.

Converse and partition. No finite null universalizes beyond its covered family. “No interaction was found” is a result about the covered family and the detector that covered it; the framework records both, so the claim cannot be quoted free of them.

Exact finite witness. Assays P3-E09 and P5-E03.

Layer-specific bridge obligations. Cluster C promotes a term Foundations V used informally, and the promotion carries two obligations. Nothing Foundations V says about enablement transfers here, because Foundations V states nothing about it that a bridge could carry — the four uses are references, not results. And the agenthood distinction between enablement and within-layer causation is inherited as a *distinction* rather than as a theorem: this paper types the distinction and proves separations within its own framework, and does not rely on the earlier treatment for any step.

Formal correspondence and boundary. This cluster corresponds to the FoundationsVII.Enablement module family together with the reachability and negative-force modules. All declarations are kernel-verified in the sense of Section 2.3.

8 Cluster D — Dynamics and residue

Six entries govern what happens when interactions are iterated, reordered, or accumulated: whether route matters, whether the result is confluent, whether an arrow exists, and what is left over. The cluster closes the paper’s body because its results depend on all three preceding clusters.

8.1 Order residue and holonomy

Theorem 8.1 (Interaction-order residue, CONDITIONAL THEOREM, VII-C017). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then two legal route results on the same typed target and members that differ in order or bracketing and disagree in predictive value define a nonzero route residue. Nonzero residue is interaction holonomy in the declared sense.*

Provenance. RECOVERED SBT (route residue relative to a predictive quotient; holonomy with memory). SHARPENED (interaction-order residue between two legal route results on the same typed target and members). *Canonical:* Foundations IV F3 Holonomy-Memory Repair Normal Form (Thm. 5.4) [33]; predictive witness (Def. 2.4), loop asymmetry (Thm. 3.6) [36]; Foundations I T-AOT-02 (Thm. 5) [30]. *Relation:* Differently scoped: the corpus defines residue for two routes on one carrier; VII compares two legal route-result records with matching target and member identifiers, changed order or bracketing, and unequal predictive values. *Lean:* FoundationsVII.RouteComparison.Holonomy, FoundationsVII.constructive_interaction_holonomy, FoundationsVII.holonomy_with_zero_arrow_witness; receipt axioms: none.

Proof of Theorem 8.1. Compare two legal route results. When their target and member identifiers agree, their order or bracketing codes differ, and their predictive values disagree, the pair has route residue and satisfies the additional predictive-value clause of holonomy by definition. The

formal development carries the route-comparison record, the residue and holonomy predicates, a constructive holonomy witness, and a holonomy-without-arrow witness. Two non-commuting interactions already admit inequivalent orderings; three or more is the first arity at which alternative bracketings arise. Neither observation is a clause of the theorem, and the finite assay records order and bracketing flags rather than an interaction count. P4-E09 enumerates 1,024 order and bracketing comparisons and populates both the zero-residue and nonzero-residue classes. $\square$

Strictness. The residue is relative to the recorded route outputs, and that relativity is essential rather than a weakening: the comparison fixes the target and member identifiers and then tests the recorded route codes and predictive values. It makes no claim about route differences outside those fields.

Holonomy here is a declared structural use. Berry’s result concerns the geometric phase acquired by a quantum eigenstate under cyclic adiabatic variation of Hamiltonian parameters [6]; no bundle, connection, curvature, or adiabatic dynamics is inferred for a VII route comparison.

Converse and partition. Holonomy does *not* imply an arrow. Order sensitivity is a statement about routes; directionality is a statement about drive. The two are separated here and kept apart by the certificate in Theorem 8.3.

Exact finite witness. Assays P4-E09, P4-E10, P5-E07 and P5-E10 enumerate the order-residue family. Within the arrow carrier a holonomy-without-arrow case is exhibited: nonzero residue, no drive certificate, no arrow credit.

8.2 Confluence and seed dependence

Theorem 8.2 (Admission confluence, CONDITIONAL THEOREM, VII-C018). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then distinct lawful admission and refinement routes are confluent exactly under the declared critical-pair and audit-equivalence conditions. Where they fail, the fixed point reached may depend on the route or on the initial seed partition.*

Provenance. RECOVERED STANDARD (vocabulary only: the critical-pair criterion for local confluence [15, 14] and the lemma that termination with local confluence gives confluence [21]). RECOVERED SBT (elementary flatness iff local confluence; odometer confluence witnesses). SHARPENED (a typed finite record over a declared admission system in which a critical-pair predicate is equivalent to the declared-scope confluence predicate, with seed dependence exhibited). *Canonical:* Thm. 5.6 (elementary flatness iff local confluence) [29]; Foundations VI G8 Odometer Abelianization (Thm. 6.1) [35]. *Relation:* Weaker and differently scoped: the standard results concern term and abstract rewriting, with termination as a hypothesis of Newman’s lemma; VII’s biconditional relates two declared-pair predicates on a finite admission system, formalises neither termination nor either standard result, and borrows only the vocabulary. *Lean:* FoundationsVII.FiniteAdmissionSystem.finite_critical_pair_criterion_iff_confluent, FoundationsVII.seed_partition_can_change_terminal_package; receipt axioms: none.

Strictness. “Exactly under” is a biconditional over the declared finite admission system — a closed family with all transitions enumerated: the conditions are not merely sufficient. Where they fail, confluence is not established, and the finite battery exhibits seed dependence (CM-23).

Converse and partition. One commuting square does not prove global confluence. A framework that accepts local commutation as evidence of confluence will report path-independence for systems that do not have it.

Exact finite witness. Assays P4-E07, P4-E08 and P5-E10.

8.3 Arrow and cross-time contact

Theorem 8.3 (Interaction arrow, CONDITIONAL THEOREM, VII-C032). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then arrow or irreversibility claims require a drive certificate recording drive, path asymmetry, failed reversal, budget and audit, distinct from order residue; the null-versus-driven controls that separate such a certificate from an artefact are exhibited alongside it. Cross-time contact claims additionally require a synchronisation or partial-order witness.*

Provenance. RECOVERED SBT (no arrow from protocol holonomy without a genuine drive; data processing for path-reversal asymmetry). SHARPENED (a typed drive certificate over interaction profiles and an explicit synchronisation / partial-order witness for cross-time contact). *Canonical:* Foundations I T-AOT-01 (Thm. 4), T-AOT-02 (Thm. 5) [30]; NG_ARROW_DPI (Thm. 3.1), NG_PROTOCOL_TRAP (Thm. 3.3) [38]; loop asymmetry (Thm. 3.6) [36]; §7 no global time from protocol holonomy [39]. *Relation:* Weaker and differently scoped: the corpus theorems are quantitative statements about Markov path laws; VII restates the discipline as a typed certificate and does not reprove the KL inequalities. *Lean:* FoundationsVII.DriveCertificate.Valid, FoundationsVII.ArrowClaim.Eligible, FoundationsVII.driven_arrow_requires_independent_drive_certificate, FoundationsVII.CrossTimeContact.Valid; receipt axioms: none.

Proof of Theorem 8.3. Suppose an arrow claim were supported by order residue alone. The formal development exhibits a claim with nonzero holonomy and no arrow credit (CM-10), and, separately, a reversible null control and a driven irreversible control that differ exactly in the presence of drive and in the equality of forward and reverse values. Residue is present in the zero-arrow claim, so residue alone does not separate a driven profile from a null; the drive certificate — drive, path asymmetry, failed reversal, budget, audit — is what an eligible arrow claim requires, and the cross-time clause follows once contact is relativized to a synchronisation or partial-order witness. P4-E10 enumerates the holonomy, drive, path-asymmetry and reversal profiles. $\square$

Strictness. The certificate demands drive, path asymmetry and failed reversal together, and none substitutes for another. The null control is what distinguishes an arrow from an artefact of the measurement, and it is exhibited rather than assumed.

Converse and partition. Order sensitivity and entropy production internal to a parent do not alone prove an inter-theory arrow. Both are the standard evidence offered for directionality; the theorem says neither suffices, because both can be present with the drive absent.

In Crooks’s setting of stochastic, microscopically reversible dynamics, forward/reverse path-probability ratios are related to entropy production [9]. That result supplies a physical comparison for the drive and reversal fields, not a physical interpretation of an abstract interaction profile absent an instantiation bridge.

Exact finite witness. Assays P4-E10, P4-E11, P5-E07 and P5-E10. In the arrow carrier the holonomy-zero-arrow case and the driven-arrow case are both populated, so the separation is exhibited from both sides.

8.4 Why the three dynamics results are ordered as they are

Holonomy, confluence, and the arrow are stated in that order because each constrains the next. Holonomy establishes that route can matter: there exist inequivalent legal orderings with nonzero residue. Confluence asks when it does not matter, and answers with a biconditional — exactly under the declared critical-pair and audit-equivalence conditions — so the two results partition the space of ordering behaviour rather than merely bounding it.

The arrow then asks a different question, and the ordering makes clear why it is different. Holonomy and confluence are about *whether* order affects the outcome; the arrow is about whether one direction is privileged. A system may have holonomy with no arrow at all, which is precisely CM-10, and a system may be perfectly confluent while being strongly driven. Directionality is orthogonal to route-dependence, and reading the third result as a strengthening of the first two is the error Theorem 8.3’s converse clause exists to block.

8.5 Residual and the needle

Definition 8.4 (Residual and dissolution ledger, FORMAL SCHEMA, VII-C019). Inaccessible, unadmitted, expired, failed, and join-created distinctions remain typed residuals. A residual entry may record its discharge route — ascent, peer join, refinement, or declared termination — and the ledger is append-only.

Provenance. RECOVERED SBT (no unstatused residual; residual status records). SHARPENED (join-created cross-term residuals typed separately from inherited and dissolved ones). *Canonical:* Foundations IV F9 No Unstatused Residual (Thm. 4.4) [33]; Foundations V E5 SixBirdsFoundationsV.ResidualStatusRecord [34]. *Relation:* Differently scoped: F9 requires every residual to carry a typed status; VII adds the join-created class and the append-only discharge ledger. *Lean:* FoundationsVII.ResidualEntry, FoundationsVII.ResidualKind, FoundationsVII.ResidualLedger.no_silent_residual_deletion_under_scope_change; receipt axioms: propext.

Converse and partition. Access growth need not monotonically reduce all obstructions. Expanding what a theory can reach may create obstructions faster than it dissolves them, and the ledger is what makes that visible instead of letting the count drift.

Exact finite witness. Assay P3-E08 enumerates 192 residual assignments over inherited, dissolved and join-created classes and accepts 11.

Theorem 8.5 (Join-created obstruction, CONDITIONAL THEOREM, VII-C036). *Assume throughout that all records are well typed and source-located; that no undeclared operation, source, or observer resource is used; and that inherited results are transported only through accepted bridge/reuse records. Then a join may dissolve parent obstructions and yet create new cross-term needles localized to the composite interface. The ledger must distinguish inherited from join-created residue.*

Provenance. RECOVERED SBT (needles as unpriced localized obstructions; no-needles stability). SHARPENED (a join-created needle localized to the composite interface, exhibited while an inherited obstruction is dissolved). *Canonical:* Foundations IV F6 No-Needles Stability (Thm. 4.1) [33]; layer-dissolving membrane theorem (Thm. 5.1) [37]; Foundations VI G6 Endogenous Needle Generation [35]. *Relation:* Differently scoped: the corpus bounds needles inside a formed package; VII shows a join can create one at the interface while dissolving parents’ obstructions. *Lean:* FoundationsVII.CrossTermNeedle.Valid, FoundationsVII.join_can_dissolve_and_create_residuals_simultaneously, FoundationsVII.joining_is_not_monotonically_obstruction_reducing; receipt axioms: propext, Quot.sound.

Proof of Theorem 8.5. Dissolution and creation are recorded in the same append-only ledger but are indexed differently: a dissolved obstruction is one inherited from a parent and discharged by a named route, while a join-created needle is indexed to the composite interface, which by construction did not exist in either parent. Since the composite interface is new, an obstruction localized there cannot be inherited, and dissolving every inherited obstruction therefore leaves the created ones untouched. The formal development exhibits the combined configuration directly: a well-formed residual transition that dissolves one inherited obstruction and creates one cross-term needle in the same step, and a before/after ledger pair in which the cross-term residual is present after the join and absent before it. P3-E08 types the three residual classes separately — inherited, dissolved, and join-created cross-term — over its declared carrier of 192 assignments, of which 11 are accepted; it enumerates single assignments rather than whole ledger states. $\square$

Strictness. This result closes a natural optimism about interaction: joining two theories is not a way of having fewer problems. The composite interface is a new place for obstructions to live, and some obstructions exist only there.

Converse and partition. Joining is not monotonically obstruction-reducing. The distinction between inherited and join-created residue is what prevents a composite from being credited with dissolving an obstruction it in fact relocated.

Exact finite witness. Assays P3-E08 and P5-E04 enumerate the residual family over their declared carriers, populating the inherited, dissolved and join-created classes separately. The enumeration establishes that the third class is populated over the carrier; the ledger-level claim that dissolution can coexist with creation is carried by the named transition witness in the formal development.

8.6 A refuted conservation law

Remark 8.6 (Contact-degree conservation, REFUTED CANDIDATE, VII-C031). The claim that contact admits a universal conserved scalar degree is **refuted** over the declared assay. What survives is the typed ledger discipline of Definition 6.8: local accounting over declared resources, with no conserved quantity asserted across settings.

Provenance. RECOVERED SBT (conservation as orbit descent). SHARPENED (the universal conserved contact degree refuted over the declared assay; typed ledgers retained). *Canonical:* Foundations IV F27 Conservation as Orbit Descent (Thm. 7.11) [33]. *Relation:* Differently scoped: F27 characterises when a quantity is conserved; VII shows no single scalar contact degree is conserved across the declared joins. *Lean:* FoundationsVII.ContactQuantityTransformation.Conserved, FoundationsVII.plausible_contact_quantities_disagree, FoundationsVII.no_unqualified_contact_degree_from_bookkeeping_alone; receipt axioms: none.

Converse and partition. The refutation does not remove the ledger; it removes the interpretation of the ledger as measuring a conserved quantity. Cross-costs, refunds and occupancy remain priced exactly as before. The data: assay P3-E11 (2,880 raw, 780 accepted); two plausible scalar degrees of one witness vector take the values 4 and 7 (FoundationsVII.plausible_contact_quantities_disagree); an apparent conservation with omitted observer occupancy is refused; CM-21 (TTW-S22, UNPRICED_OBSERVER) and CM-25 (TTW-S24, REFINEMENT_DESTROY_JOIN).

Layer-specific bridge obligations. Cluster D specialises Foundations VI’s dynamical vocabulary, and two obligations follow. An application of an inherited quotient-relative holonomy result must identify its quotient readout with the predictive-value field of VII’s route-result records; VII’s formal statement itself contains no quotient object. And the arrow certificate of Theorem 8.3 is stated over inter-theory interaction, whereas the inherited directionality results concern dynamics internal to one package; the theorem’s own converse clause records that the latter do not imply the former.

Formal correspondence and boundary. This cluster corresponds to the `FoundationsVII.Dynamics` and `FoundationsVII.Residuals` module families. All declarations are kernel-verified in the sense of Section 2.3; per Remark 2.1 this does not alter the disposition of any entry above.

9 Cross-family corollaries and the null battery

The four clusters are stated independently, but their results interact. This section records the twenty-one corollaries that hold across cluster boundaries, and the null battery that keeps them honest.

9.1 Why cross-family results need their own section

A corollary that draws on Cluster B and Cluster D at once inherits the hypotheses of both. Stating it inside either cluster would hide half its assumptions in the other section, so the cross-family results are collected here with their spanning candidates named; each corollary’s full hypothesis set lives in its Lean statement, located via Section A. Twenty-one such corollaries are recorded; Section A tabulates each with its supporting entries.

The twenty-one corollaries fall into nine families, each named for the clusters it spans. They come in a characteristic shape: within each family, one corollary states what a certified configuration *gives* you, and one or two state what it *still does not*. That pairing is deliberate — a cross-family result that only reported its positive half would be the easiest place in the paper for scope to leak.

Prospective join (FVII-COR-001–FVII-COR-003). Spanning VII-C002, VII-C006, VII-C009, VII-C011. A certified configuration carries a temporal source, a join, and payment together (FVII-COR-001); retrospective registration blocks prospective join credit (FVII-COR-002); and certification preserves declared source alignment (FVII-COR-003). The three certificates do not collapse into one: timing, admission, strictness, alignment and payment remain separately discharged.

Source and budget (FVII-COR-004–FVII-COR-005). Spanning VII-C009, VII-C010, VII-C011, VII-C029. Certification yields independence, payment, and a capacity bound together (FVII-COR-004), and carries the source gate and the payment gate as two separate conjuncts (FVII-COR-005): each is discharged on its own, and the certificate records both rather than inferring one from the other.

Non-interaction (FVII-COR-006–FVII-COR-007). Spanning VII-C008, VII-C033. Certified non-interaction excludes a strict join within the covered family (FVII-COR-006) while *retaining* the named escape routes outside it (FVII-COR-007). The second is what keeps the first from being read as a negative that extends past its covered family.

Enablement residual (FVII-COR-008–FVII-COR-009). Spanning VII-C019, VII-C027, VII-C036. A certified chain accumulates resources and preserves residuals (FVII-COR-008); a join-created needle blocks the zero-residual reading (FVII-COR-009). A chain that appears to have discharged everything has usually relocated something.

Refinement descent (FVII-COR-010–FVII-COR-012). Spanning VII-C015, VII-C030, VII-C034. A preserving refinement with a valid descent certificate preserves all three join gates (FVII-COR-010); a destroying refinement exhibits a failed gate (FVII-COR-011); and a valid descent certificate does not erase a refinement obstruction (FVII-COR-012). This family is where Remark 6.5’s four-way classification does its work.

No-free-join (FVII-COR-013–FVII-COR-014). Spanning VII-C003, VII-C011, VII-C035. The first join in a closed regime faces a two-gate obstruction — bootstrap and payment simultaneously (FVII-COR-013) — and the escape requires both a seed and payment (FVII-COR-014). Satisfying one gate does not open the other.

Holonomy and arrow (FVII-COR-015–FVII-COR-016). Spanning VII-C009, VII-C017, VII-C032. Strict join and holonomy together do not replace a drive certificate (FVII-COR-015); holonomy together with a valid drive certificate and a recorded arrow credit is an eligible arrow claim (FVII-COR-016). The pair fixes what holonomy contributes: it never substitutes for the drive certificate, and an eligible arrow claim rests on that certificate rather than on route residue.

Negative force (FVII-COR-017–FVII-COR-018). Spanning VII-C021, VII-C023, VII-C033. A certified negative excludes contact and occurrence at the declared scope (FVII-COR-017); a point null plus a bounded horizon is not automatically global (FVII-COR-018).

Observer endogeny (FVII-COR-019–FVII-COR-021). Spanning VII-C013, VII-C029. Certification requires pricing and internal execution (FVII-COR-019); an unpriced observer blocks joint native endogenous credit (FVII-COR-020); and the zero-occupancy escape preserves endogenous credit (FVII-COR-021). The third is the escape route of NGVII-05, stated as a positive result rather than left implicit in the obstruction.

9.2 What the corollaries do not license

Each family carries an explicit nonclaim, and the nonclaims are the load-bearing part of this section. Stated together:

- **Prospective join.** Prospective timing, lawful admission, strictness, source alignment, and payment remain *separate* certificates; none is inferred from the others.
- **Source and budget.** Finite capacity is a declared local resource bound, not a conserved interaction currency.
- **Non-interaction.** The exclusion applies only inside the exact covered family, and preserves the named outside-family escape routes.
- **Enablement residual.** Residual debt is typed bookkeeping for the declared chain; it does not establish a scalar residual law.
- **Refinement descent.** A valid descent certificate does not repair a refinement that destroys compatibility or strictness.

- **No-free-join.** The obstruction is relative to the declared seed, generator, external-provision, payment, and certified-zero-cost channels.
- **Holonomy and arrow.** Strict join and holonomy do not supply directionality without an independent drive certificate.
- **Negative force.** Negative force is coverage-, detector-, budget-, family-, and horizon-qualified; bounded nulls are not globalised.
- **Observer endogeny.** Endogenous credit does not mean environmentally isolated or uncaused, and observer pricing remains a separate obligation.

9.3 The null battery

The positive results are accompanied by null fixtures in the reference world: cases in which a law’s antecedent is withheld and whose expected status records that the credit is refused. The detector contract of Theorem 5.9 records non-empty null, falsifier and control case lists over the frozen fixtures; it does not pair each law with a specific null, and the pairing below is the paper’s reading of the atlas. A law that fires when it should not is as defective as one that fails to fire, and the battery is what detects the first kind.

The battery matters most for the negative results. A no-go that fires on every input is not evidence of an obstruction; it is evidence of a broken detector. Each of the eleven fronts is therefore accompanied by its named escape routes, enumerated over the declared carrier, so that both the firing and non-firing cases are populated.

What a null looks like here. A null is not merely an absence of signal. It is a configuration in which the antecedent of a law is deliberately withheld and the conclusion is checked to fail — and it must fail for the *stated* reason, not incidentally. Four kinds recur.

A *same-source* null pairs two packages sharing a lineage and checks that an independence-sensitive credit is refused. A *no-contact* null pairs two packages with no witnessed contact and checks that no join status above “no contact” is issued. A *scheduling* null reorders an interaction without changing its content and checks that no joint novelty is credited — CM-04 is its named witness. A *relabelling* null renames without restructuring and checks the same, with CM-14 as witness.

These four are exactly the controls the detector contract of Theorem 5.9 requires, and they are chosen because they are the four cheapest ways to produce an apparent interaction. A law that fires on any of them is mis-specified rather than strong.

Asymmetric costs. The battery treats false positives as more expensive than false negatives, and the asymmetry is deliberate. A missed interaction leaves a question open; a claimed interaction that is not there enters the literature as a result and is built upon. The detector contract encodes this by recording a false-positive cost above the false-negative cost and by listing the four control families as non-empty case sets; the rule that the controls must pass before a positive is issued is the discipline this paper applies to the atlas, not a clause of the contract, while a negative requires only that its coverage be declared.

What the battery does not do. It cannot detect a law that is correct but uninteresting, nor one whose carrier is too small to contain the case that would refute it. A null battery establishes that the laws discriminate within the declared family; the question of whether the family is the right one is answered by the argument, not the enumeration.

10 The Two-Theory World and finite laboratories

The abstract catalog is anchored by one concrete finite model, exhaustively enumerated. This section describes it and what it is for.

10.1 The reference world

The Two-Theory World is a finite reference model containing exactly two theory packages and the machinery for them to interact: access records, admission transitions, contact surfaces, join entries, enablement chains, budget ledgers, and audits. It is small enough to enumerate completely over its declared carrier and rich enough that every law in Sections 5 to 8 has an instance in it.

Twenty-four frozen scenarios and twenty-seven named countermodels are declared against it. The scenarios are positive: each exercises a law and is expected to produce a specific status. The countermodels are negative: each is constructed to refute a specific named claim, and is expected to be rejected for a specific stated reason.

10.2 What a frozen fixture buys

The scenarios and countermodels are frozen: each fixture carries its expected status and its recorded hash, so a fixture whose expected value is changed to match a result changes its hash. This is what distinguishes a test from a description. A fixture whose expected value can be adjusted silently after seeing the result tests nothing.

Two independent implementations evaluate the same fixtures — one in Lean, one in Python — and their outputs are compared exactly. Agreement is not evidence that the laws are correct; it is evidence that the two implementations agree about what the laws say, which is a weaker and more honest claim. A shared misunderstanding would survive the comparison; the proofs establish only that the recorded statements follow from their hypotheses, and whether the definitions capture the intended notion is argued in the paper, not checked (Section 14).

10.3 The bounded carriers

Beyond the reference world, each cluster declares finite carriers for exhaustive enumeration. Forty-three assays are recorded across four phases. Summed over all of them the carriers comprise 284,524 raw cases reducing to 214,636 canonical, with 70,915 accepted and 143,721 rejected.

The aggregate is less informative than its structure, so the families are given in full below. Three features of the distribution can be read off directly.

First, the carriers are *deliberately uneven*. The strictness family (P3-E03) alone accounts for 131,072 of the 284,524 raw cases, because it enumerates omission attacks over sixteen independently represented certificate coordinates and every omission must be checked. Most families are two to three orders of magnitude smaller, because the property under test has far fewer degrees of freedom. A family’s size measures the dimensionality of what is being checked, not its importance.

Second, *canonicalization applies where a declared symmetry is a genuine symmetry of the property, and nowhere else*. P2-E05 halves under source-ID renaming; P3-E02 and P3-E03 halve under left/right exchange of the two packages; P5-E10 reduces under domain permutation. In the remaining families raw and canonical coincide and no reduction is claimed — an important discipline, since an unjustified quotient would shrink a carrier without shrinking the space of things that can go wrong in it.

Third, the accepted and rejected counts are *implementation receipts*. They record what each declared carrier contains under its law and let a reader re-run the enumeration; they are not evidence for or against any law, and no acceptance rate is read as such in this paper.

Phase-2: 9 assays

Assay	Raw	Canon.	Acc.	Family
P2-E01	128	128	14	Seven-coordinate access normal forms satisfying only the declared implication spine.
P2-E02	128	128	12	Operational profiles separating rule soundness, execution, reachability, firing, and occurrence.
P2-E03	32	32	29	Bootstrap assignments with admitted-seed, reachable-generator, and open-family external-provision authorization.
P2-E04	256	256	129	Prospective-credit assignments with preregistration, precedence, task blindness, outcome independence, symmetry, and matched controls.
P2-E05	512	256	128	Two-package provenance/access profiles normalized by source-ID renaming.
P2-E06	32	32	19	Total-lens transfer assignments with self-owned targets and explicit partial-target adapter licensing.
P2-E07	64	64	33	Finite-horizon negative-force assignments with family closure and detector-power gates.
P2-E08	18	18	15	Observer occupancy/pricing assignments over costs 0,1,2.
P2-E09	162	162	91	Failed-admission settlement assignments over costs 0,1,2.
Total	1,332	1,076	470	

Phase-3: 11 assays

Assay	Raw	Canon.	Acc.	Family
P3-E01	256	256	54	Compatibility, evidenced contact, peer transport, composite formation, and strict-join claims.
P3-E02	4,096	2,048	170	Declared-family join evidence profiles, canonicalized under left/right parent exchange.
P3-E03	131,072	65,536	32,770	Strict-join omission attacks over independently represented certificate axes.
P3-E04	128	128	72	Source-lineage and independence-sensitive credit assignments.
P3-E05	432	432	210	Typed join payment, refund, observer occupancy, and certified-zero-cost assignments.
P3-E06	200	200	100	Finite live-join capacity profiles over capacities and positive unit costs up to four.
P3-E07	576	576	138	Parent-retention and refinement effects with preserving, strengthening, weakening, and destroying cases.
P3-E08	192	192	11	Inherited, dissolved, and join-created residual assignments with typed cross-term provenance.
P3-E09	256	256	128	Coverage-qualified non-interaction profiles with an explicit outside-family escape marker.
P3-E10	1,024	1,024	99	Conditional product, pullback, and pushout representations of strict join.
P3-E11	2,880	2,880	780	Candidate contact/join quantities over bounded values with exact scoped conservation credit.
Total	141,112	73,528	34,532	

Phase-4: 12 assays

Assay	Raw	Canon.	Acc.	Family
P4-E01	1,792	1,792	917	Source-typed enablement attribution profiles with honest bridge-refinement stability.
P4-E02	2,048	2,048	1,024	Exact carried/reachable/executed/audited/budgeted endogenous-credit profiles.
P4-E03	1,280	1,280	160	Closure-agency and relation/layer/participant-birth profiles with objecthood gates.
P4-E04	6,144	6,144	1,605	Upward/downward/peer transmission profiles with payload, loss, ambiguity, source, budget, and causal-certificate gates.
P4-E05	512	512	128	Constructive enablement/descent and necessary/sufficient separation profiles.
P4-E06	41,472	41,472	10,449	Two-link enablement composition with source/budget/residual/audit compatibility and accumulated cost/debt coordinates.
P4-E07	64	64	32	Finite critical-pair profiles distinguishing joinability, audit equivalence, terminal equality, and predictive equivalence.
P4-E08	64	64	21	Seed-partition dependence profiles separated from presentation-only route differences.
P4-E09	1,024	1,024	256	Typed order/bracketing comparison profiles defining route residue and predictive holonomy on one target.
P4-E10	128	128	64	Holonomy/drive/path-asymmetry/reversal profiles with an independent arrow certificate.
P4-E11	384	384	96	Cross-time contact via synchronization or partial-order witnesses with admissible reparameterization controls.
P4-E12	256	256	80	Primitive-operation algebra readiness profiles plus a separately scoped resource-delta fragment.
Total	55,168	55,168	14,832	

Phase-5: 11 assays

Assay	Raw	Canon.	Acc.	Family
P5-E01	8,192	8,192	4,096	Prospective commitment, lawful admission, strict-join, source, and payment integration.
P5-E02	3,456	3,456	864	Source-independent, observer-priced strict-join credit and finite live-capacity assignments.
P5-E03	512	512	255	Coverage-qualified certified non-interaction and strict-join exclusion.
P5-E04	18,432	18,432	4,826	Residual-aware two-link enablement composition with append-only retention and cross-term debt.
P5-E05	2,048	2,048	496	Parent-refinement effects composed with source-, budget-, and square-preserving descent fidelity.
P5-E06	4,096	4,096	256	Combined bootstrap and payment assignments for first-join credit.
P5-E07	1,024	1,024	400	Strict-join, route-holonomy, and independently driven arrow assignments.
P5-E08	4,096	4,096	288	Operational reachability plus horizon-, closure-, and detector-qualified negative force.
P5-E09	18,432	18,432	2,304	Observer-pricing and declared-boundary endogeny assignments.
P5-E10	8,192	6,144	1,152	Three-domain critical-pair, seed-partition, and interaction-order assignments.
P5-E11	18,432	18,432	6,144	Integrated two- and three-domain release profiles with typed sources, budgets, observers, joins, residuals, and scoped negatives.
Total	86,912	84,864	21,081	

10.4 The countermodel atlas

Twenty-seven named countermodels are declared against the reference world. Each is constructed to refute one named claim, and each is rejected for one stated reason. They are listed here in full, because a countermodel whose content is left in a data file is one a reader cannot dispute.

Origin, carrier and contact. CM-01 same source but no shared access: common origin does not imply shared access or independence. CM-19 shared carrier without lawful peer transport: co-location does not provide an executable bridge. CM-02 common interface without witnessed contact: compatibility implies neither contact nor non-interaction. CM-12 total-lens result failing on partial self-owned access: totality does not transfer silently. Together these four close the inference from *sharing something* to *interacting*, which is the most common way an interaction claim is made without evidence.

Contact, join and fake joins. CM-03 contact without strict joint information: contact does not imply strict join. CM-27 lawful parents with no common admissible package: parent lawfulness does not guarantee that a join exists — the countermodel behind Definition 3.5’s conditional existence. CM-13 product or common refinement without strictness. CM-14 relabelling-only fake join: presentation change cannot manufacture strict content. CM-04 scheduling-only apparent joint novelty: post-hoc scheduling can manufacture an apparent joint signal. CM-26 one-lineage pseudo-join: behavioral difference inside one lineage does not establish source independence.

The countermodels recorded against Theorem 6.10 are a different three: CM-03 contact without strict joint information, CM-19 shared carrier without lawful peer transport, and CM-21 a zero-cost join hiding positive observer cost. Two of the theorem’s clauses have a direct witness among them — CM-19 for the first-capability clause and CM-21 for unpaid formation — while CM-03 bears on the family generally rather than on a single clause. The relabelling clause’s own witness is CM-14, which the registry records against VII-C007 and VII-C009 rather than against VII-C035; no one-to-one correspondence between the three clauses and these three countermodels is claimed. CM-04 and CM-26 bear on strictness and source independence instead.

Retention and refinement. CM-05 a join that loses one child: composite formation does not imply parent retention. CM-22 parent refinement destroying an admissible join: join admissibility is not monotone under refinement — the witness behind Remark 6.5’s destroy class. CM-15 strict extension without certified objecthood: non-factorization does not imply closure.

Enablement and descent. CM-06 enablement without descent: an enabling condition can be load-bearing without factorization. CM-07 enablement necessary but insufficient: an enabler may be necessary while alternatives still determine the outcome. Together these two are the witness of Theorem 7.2. CM-17 theorist-triggered enablement: execution by the analyst fails the carried-generator condition, which is the negative control for Theorem 7.3.

Reachability and occurrence. CM-08 sound but unreachable admission rule: soundness does not imply operational reachability. CM-09 reachable admission that never occurs: reachability does not imply occurrence. CM-18 finite negative search with an open family: a bounded null does not prove a universal no-go — the discipline of Definition 7.8 in one witness.

Order, arrow and residue. CM-10 holonomy with zero arrow: route residue does not imply directionality. CM-16 strict join without directionality: strict novelty does not imply an arrow. CM-24 identical members assembled in different orders: bracketing may change the audited result. CM-23 seed-dependent closure fixed points: the initial partition can change the terminal package, which is Theorem 8.2’s failure case. CM-25 a join manufacturing a cross-term needle: joining need not monotonically dissolve obstructions, the witness behind Theorem 8.5.

Observer and cost. CM-11 observer occupancy omitted from budget: hidden observer resources defeat endogenous credit. CM-21 zero-cost join hiding positive observer cost: unpriced occupancy invalidates a no-cost claim. CM-20 post-hoc prediction self-certification: retrospective admission cannot earn prospective credit, the witness for NGVII-03.

What the atlas is for. Read together, the twenty-seven are a map of the ways an interaction claim can look supported and not be. Most refute an inference from *something shared* or *something observed* to a joint object, or from a structural property to a stronger one; the rest separate credits that are routinely conflated — strictness from direction, reachability from occurrence, enablement from descent, dissolution from relocation. A framework that admitted any one of these inferences would be able to certify a join that is not there.

Remark 10.1 (The boundary of finite evidence). Exhaustive enumeration over a declared carrier establishes the status of every case inside that carrier — accepted or rejected under the law — and nothing about cases outside it. Where a law is stated for all packages satisfying its hypotheses, the

warrant is the proof; the enumeration is a control against the proof having assumed something its hypotheses do not supply.

Remark 10.1 is why every finite figure in this paper is quoted with its carrier, and why no result is stated as holding generally on the strength of an enumeration alone.

11 Layer instantiations at a glance

The catalog is layer-agnostic: nothing in Sections 5 to 8 is specific to physics, mathematics, biology, or society. This section indicates where the laws are expected to bite, and marks the status of each indication honestly. The laws invoked below are the entries of Sections 5 to 8 with the provenance printed there; no indication adds a claim to them.

11.1 How to read this section

Every entry below is a *pointer*, not a result. An instantiation is a claim that a particular real setting satisfies the hypotheses of a particular law, and establishing it requires supplying the witnesses that law demands — the same witnesses the catalog demands everywhere else. No instantiation in this section has been carried out; they indicate where the work would go.

This is the section most likely to be quoted out of context, so the constraint is stated once and absolutely: nothing here licenses applying a law to a setting without first supplying that setting’s own witnesses. An instantiation is not an application; it is a hypothesis about where an application might succeed.

11.2 Indicative settings

Two formal theories with a shared model class. The most direct setting. Cluster B applies where two axiomatisations are candidates for combination, and the questions the catalog asks are the ones that are usually skipped: does a common model class constitute a join (Theorem 6.2 says that even certified transport does not by itself form one), does the combined theory retain both parents recoverably, and does it carry an operation factoring neither through either parent nor through their declared product or common refinement (Theorem 6.4). *What must be supplied:* the admissibility regime fixing which combinations are lawful; the comparison baseline against which non-factorization is judged — without it the anti-product witness has no content; and a budget, since Definition 6.8 prices interface access and joint novelty even where the cost is nominal.

Instrument-mediated measurement. Theorem 6.9 bears on any setting where apparatus consumes shared capacity. The law does not forbid the instrument; it forbids certifying native or endogenous formation while omitting the instrument’s occupancy, source role, or cost. *What must be supplied:* the declared capacity and the instrument’s draw on it; whether the observer is external (NGVII-05’s first escape) or genuinely zero-occupancy (its second, which must be certified rather than assumed). The failure mode this catches is a system credited with forming something by itself while the apparatus that made the formation possible is not on the books.

Layer formation in composite systems. Cluster C applies wherever a lower layer is said to make a higher one possible. The relevant test is the hidden-executor clause of Theorem 7.3: if the enabling step is performed by the analyst rather than carried, reached and executed within the declared boundary, the system has been assembled rather than formed. *What must be supplied:*

the boundary itself, since endogeny is boundary-relative and enlarging the boundary changes what counts as carried, reached and hidden, so the boundary must be declared before the criterion is applied; the enablement records with their attribution fields; and the budget, since endogenous formation must be paid for from inside. Note that accounted environmental coupling is compatible with endogeny — what is not is unaccounted coupling.

Sequential interventions. Cluster D applies wherever interventions are applied in a choosable order and outcomes are compared across orderings. Theorem 8.1 gives the residue; Theorem 8.3 says the residue is not yet an arrow. *What must be supplied:* two legal route-result records with matching target and member identifiers, their order and bracketing codes, and their predictive values; and, for any directionality claim, a drive certificate with null-versus-driven controls. The failure mode here is treating order-sensitivity as evidence of an arrow, which CM-10 exhibits directly.

11.3 What would have to be supplied

Common to all four: a declared admissibility regime; a declared carrier or interface family; the specific witnesses the invoked law names; and an audit. Absent those, a setting is a place where the question can be asked, not a place where it has been answered.

Two further cautions are worth stating, because they are the ways an instantiation goes wrong while appearing to succeed. First, the declared family must be fixed *before* the evidence is examined; a family narrowed after the fact to exclude inconvenient cases is not a declared family, and NGVII-03 is the front that catches this. Second, a negative result obtained in an instantiation inherits the force of its search family and no more (Definition 7.8): finding no interaction in a setting establishes something about that setting’s covered family, not about the theories in general.

12 Deferred candidates and future work

Two candidates are closed as deferrals. A deferral in this framework is not an open problem left for a later paper; it is a boundary drawn deliberately, with the landed fragment named and the deferred part stated. Recording them as closed rather than open is what prevents the paper from appearing to claim territory it has not taken.

12.1 Categorical reduction, CLOSED DEFERRAL, VII-C026

The question is whether the join of theories reduces to a categorical construction — a product, pullback, or pushout — under explicit hypotheses.

Remark 12.1 (Categorical reduction, CLOSED DEFERRAL, VII-C026). Closed as a deferral. The unconditional reduction is not established, and this paper does not pursue it. The landed fragment is the negative direction: the anti-product witness of Theorem 6.4, which shows that a strict join carries an operation factoring neither through either parent nor through the declared product or common refinement, and therefore cannot be a product in the relevant sense; and the conditional form, under which a special-case representation is permitted only with a separately certified anti-product witness.

Provenance. RECOVERED STANDARD (products, pullbacks and pushouts as limit and colimit constructions [18]). SHARPENED (scope repair: unconditional reduction of the join to a limit or colimit construction rejected, special cases permitted under an explicit category). *Canonical:* Foundations IV F51 Unification as Common Refinement (Thm. 8.18) [33]. *Relation:* Differently scoped: the standard constructions are cited

for their definitions only; nothing about them is reproved. *Lean*: `FoundationsVII.UniversalConstructionCertificate.RepresentsStrictJoin`, `FoundationsVII.mere_product_is_not_strict_join_credit`, `FoundationsVII.special_case_categorical_representation_is_conditional`; receipt axioms: `propext`, `Quot.sound`.

Institution theory and algebraic specification make signatures, sentences, models, satisfaction, and morphisms explicit, while categorical constructions require an ambient category and specified arrows [12, 25]. Work on Morita and categorical equivalence likewise shows that an equivalence criterion encodes choices about preserved structure [3, 41]. Together these sources identify data that the reopen conditions must fix; they do not settle those conditions.

What remains permitted. Special-case representations under explicit hypotheses are not excluded. A setting in which the join does coincide with a categorical construction is admissible; what is refused is the unconditional identification, imposed by fiat rather than proved.

12.2 Primitive-operation algebra, CLOSED DEFERRAL, VII-C028

The question is whether the primitive operations of the interaction calculus admit a full generators-and-relations presentation.

Remark 12.2 (Primitive-operation algebra, CLOSED DEFERRAL, VII-C028). Closed as a deferral. The full algebra is not landed. The landed fragment is the resource-delta fragment: cost and residual-debt deltas combine associatively with a two-sided identity, which is what the enablement-chain composition of Cluster C requires and no more; the reopen condition for the full programme is recorded formally.

Provenance. ELEMENTARY (the resource-delta fragment is a monoid: associative combination with a two-sided identity). SHARPENED (scope repair: the full generators-and-relations algebra deferred with a formal reopen condition). *Canonical:* none beyond the elementary monoid laws. *Relation:* Not a recovered result; the landed fragment is elementary and the deferred part is stated, not claimed. *Lean*: `FoundationsVII.AlgebraReadiness.Ready`, `FoundationsVII.full_generators_relations_program_deferred_with_formal_reopen_condition`, `FoundationsVII.resource_delta_fragment_is_associative`; receipt axioms: none / `propext`.

What remains permitted. The fragment is sound as far as it goes, and nothing in the paper depends on the full algebra existing. A later presentation would extend the fragment rather than replace it.

12.3 Why these are deferrals and not gaps

The distinction matters enough to state directly, because the two look identical from outside. A gap is a place where the framework needs an answer and does not have one: something downstream depends on it, and the dependency is unmet. A deferral is a place where the framework has decided the question is out of scope, and nothing downstream depends on the answer.

Both entries above are the second kind, and the test is mechanical. Remove VII-C026 entirely and every other result in this paper still stands, because no result invokes a categorical reduction — the anti-product witness of Theorem 6.4 is stated positively, in terms of what a strict join must carry, rather than negatively in terms of what it is not. Remove VII-C028 and the composition results still stand, because Theorem 7.6 needs only that resource deltas combine associatively with an identity, which is the landed fragment; Cluster B’s payment ledger does not depend on the fragment at all.

A deferral that failed this test would be a gap wearing a deferral’s label, and the framework would be claiming closure it has not achieved. Recording both as CLOSED DEFERRAL rather than as open problems is therefore a checkable claim, not a presentational choice.

12.4 Future work

The two deferrals are the natural next objects. Beyond them, three directions are indicated by results in this paper rather than by ambition: completing the status calculus of Definition 6.3 to a genuine completeness theorem; extending the confluence conditions of Theorem 8.2 beyond the declared critical-pair family; and instantiating the catalog in a setting where the witnesses can actually be supplied, per Section 11.

13 Nonclaims and caveats

Every entry in the catalog carries recorded nonclaims, stated with the entry where it appears. This section collects the ones that apply to the paper as a whole. It is not a disclaimer; the boundaries below are as much a result of the work as the laws are, and several of them were established by refutation rather than assumed out of caution.

13.1 On grades

No result in this paper is proved by the Lean kernel in the sense of having its mathematical grade established there. The kernel confirms that the recorded declarations elaborate and depend on the axioms stated in Section 14. Fifteen entries are FORMAL SCHEMA; they define and record, and prove nothing. Fifteen are CONDITIONAL THEOREM and hold under the hypotheses stated in their own statements. The remainder are refutations, deferrals, a countermodel, and one finite decidability result. Per Remark 2.1, kernel verification does not move an entry between these classes.

13.2 On finite evidence

The finite results are exhaustive over declared bounded carriers and over nothing else. Summed over all 43 assays the carriers comprise 284,524 raw cases reducing to 214,636 canonical; the eleven Phase-5 closure assays together account for 86,912 of that raw total, and the largest single assay is the strictness family at 131,072 raw. No finite null in this paper universalizes beyond its covered family, and where a null is quoted its coverage, detector power, budget, horizon, and search-family closure travel with it.

13.3 On the join

The join of theories is not Foundations V’s repair join, is not a lattice upper bound, and is not a categorical product. Its existence is conditional and costed. A well-formed join-entry record does not establish that a join exists, and a strict join does not supply directionality.

13.4 On negative results

The eleven no-go fronts are scoped obstructions with named escape routes. Quoting one without its escapes misstates it. In particular the bootstrap obstruction (Theorem 5.4) applies to a *declared closed* family and permits admitted seeds, reachable generators, and open external provision; the

no-free-join family (Theorem 6.10) permits admitted seeds, declared bridges, and proven zero-cost channels.

Two candidates are refuted rather than established, and both retain conditional content: unconditional refinement monotonicity is false while the four-way typed classification stands (Remark 6.5), and a universal conserved contact degree is false over the declared assay while typed ledgers remain controlling.

13.5 On instantiation

No instantiation has been carried out. Section 11 indicates settings where the laws are expected to bite; each would require its own regime, carrier, witnesses, and audit before any law could be applied to it.

13.6 On what would refute this paper

A framework that could not be wrong would not be worth stating, so it is worth being explicit about what would refute the results here.

Any `CONDITIONAL THEOREM` is refuted by a configuration satisfying its hypotheses and failing its conclusion. Because the hypotheses are stated inside the statements, such a configuration can be constructed without interpretive dispute — the three standing hypotheses are checkable properties of a record, not judgements.

Any obstruction is refuted by a lawful route to the forbidden credit that is not one of its named escapes. This is the most likely refutation to succeed, and the escapes are enumerated precisely so that a candidate route can be checked against them rather than argued about.

Any finite claim is refuted by a case inside its declared carrier that the enumeration missed, which is checkable by re-running the enumeration; the carriers and their hashes are published for this reason.

Any `FORMAL SCHEMA` cannot be refuted in this sense, because it asserts nothing. It can be shown *inadequate* — by exhibiting an interaction it cannot record — which is a different and weaker kind of failure, and the one most likely to apply here.

13.7 On scope

This paper catalogs laws governing when interaction between theories is lawful. It does not supply the interaction calculus itself, which is Foundations III's; it does not establish that any particular pair of real theories interacts; and it does not claim that the catalog is complete. The status calculus of Definition 6.3 is explicitly not asserted to be exhaustive outside its declared interface family.

14 Lean formalization and finite validation

The catalog is accompanied by a machine-checked formal development. This section states precisely what was checked, what the check establishes, and what it does not.

14.1 The surface

The public development comprises 1,477 declarations across 94 declaration-bearing modules (111 public modules in total, the remaining 17 being import-wrapper modules that declare nothing): 732 theorems, 484 definitions, 192 structures, 51 inductive types, and 18 abbreviations. It is built on `leanprover/lean4:v4.28.0`, whose Lean 4 architecture combines a relatively small trusted kernel

with an extensible elaborator and programming language [11], and is *Mathlib-free*: no external library is fetched, so the trust base is the Lean kernel together with the vendored Foundations I–Foundations VI scaffolds, and the build is reproducible from the pinned toolchain alone.

14.2 The axiom receipt

An executed `#print axioms` was captured for every one of the 732 public theorems. The census:

Axiom dependency	Theorems	Share (%)
no axioms	625	85.4
<code>propext</code>	97	13.3
<code>propext</code> , <code>Quot.sound</code>	10	1.4

Two facts are worth stating plainly. There is no `sorryAx` anywhere in the 732: no hole, no admitted step. And there is no `Classical.choice`: the only axioms reached are `propext` and `Quot.sound`, two of Lean’s standard axioms [17] rather than axioms introduced by this project; the 97 and 10 theorems whose receipts name them do depend on them. `sorryAx` is the placeholder axiom Lean attaches to an unfinished proof, and its absence is what “no hole” means.

Remark 14.1 (What the receipt is not). It would be wrong to describe this as 732 axiom-free theorems. 625 are axiom-free; 97 depend on `propext`. The distinction is recorded per theorem rather than aggregated, and the release validation fails if any theorem’s recorded classification disagrees with its own receipt.

14.3 What is formalised, and what is not

The development formalises the *objects and their laws*, not the informal argument around them. Concretely: the seven access coordinates and their implication spine; the five admission transitions with their witness requirements; the contact modes and their licensing conditions; the join entry, status calculus, strictness certificate and budget ledger; the enablement record with its attribution fields; the transmission records; the order-residue comparison; and the residual ledger. Each is a structure or definition with its stated properties proved as theorems over it.

What is not formalised is the claim that these are the right objects. Definition 3.5 could be replaced by a different definition and the development would still compile, with different theorems. The kernel checks internal consequence; the argument that these definitions capture lawful theory interaction is the paper’s.

Nor is the finite evidence a proof of the general laws. The enumerations classify every case *in a declared carrier* as accepted or rejected under the law and record the counts; the proofs establish the law under its hypotheses. Where both exist for one entry they are independent checks, and their agreement is worth more than either alone precisely because they fail differently: a proof can assume something its hypotheses do not supply, and an enumeration can run over a carrier too small to contain a counterexample.

14.4 Cross-implementation differential

Independently of the axiom story, every Lean finite runner is checked against the Python laboratory. Five differentials are performed — one per phase — comparing raw, canonical, accepted, and rejected cardinalities across the 24 frozen scenarios, the 27 named countermodels, and all declared envelopes. All five agree exactly.

As noted in Section 10, agreement between two implementations is evidence that they read the laws the same way, not that the laws are correct. It is the weaker claim, and it is the one made.

14.5 Reproduction

The entire pipeline is one command, which regenerates the Python baselines, performs a clean kernel build, executes the five finite runners, runs the differentials, captures the axiom receipt, and re-validates. The bounded carriers are checked by kernel evaluation, so the build is multi-minute per envelope module and requires an enlarged worker stack; both are recorded in the build configuration rather than left to be discovered.

14.6 What the formalization establishes

Four things, and no more. That the recorded statements are well formed — they parse, typecheck, and mean something definite. That they elaborate — every proof is complete, with no hole anywhere in the 732. That their proofs depend on exactly the axioms listed in Section 14.2, neither more nor fewer. And that the Lean and Python implementations agree on every finite fixture.

It does not establish that the statements are the right ones. That is a question about whether the definitions capture what they intend to capture, and no proof assistant answers it. Nor does it establish that the catalog is complete, that the carriers are the relevant ones, or that a law is interesting. A development can be flawlessly checked and thoroughly beside the point, and the checking cannot tell the difference.

This distinction is part of the wider trust problem for machine-checked proof: kernel checking narrows what must be trusted, but confidence still depends on the statement presented to the kernel and on the surrounding implementation and review process [24].

The honest summary is that machine checking removes one class of error — the class where an encoded Lean statement does not follow from its encoded hypotheses — and leaves every other class untouched. It is worth having for exactly that reason, and worth describing precisely for the same one.

15 Promotion, inventory, and validation audit

This section records what was closed, at what grade, and under what checks. It exists so that the claims of the preceding sections can be counted and audited rather than taken on the paper’s word.

15.1 Inventory

Artefact	Count
Candidates closed	36
Objects closed	18
Formalization targets closed	20
No-go fronts	11
Cross-family corollaries	21
Decision rulings	15
Finite assays	43
Named countermodels	27
Minimized bounded rejection witnesses	11
Inherited adapter contracts	30

15.2 Provenance census

Over the 36 entries: RECOVERED SBT 34, SHARPENED 36, ELEMENTARY 3, RECOVERED STANDARD 2, NEW 1 (Section D). Over the 27 countermodels: SHARPENED 27, RECOVERED SBT 22. All 21 corollaries are SHARPENED.

15.3 Disposition census

Disposition	Count
CONDITIONAL THEOREM	15
FORMAL SCHEMA	15
REFUTED CANDIDATE	2
CLOSED DEFERRAL	2
CONSTRUCTIVE COUNTERMODEL	1
LEAN DECIDABLE FINITE	1
Kernel-proved	0

The final row is printed rather than omitted. No entry in this paper carries a disposition of kernel-proved, and a reader should be able to see that the count is zero rather than infer it from silence.

15.4 Immutability of inherited surfaces

The vendored Foundations V and Foundations VI libraries are checked byte-identically against their baseline on every validation run. No result in this paper is obtained by editing an inherited file, and the check is mechanical rather than promised.

15.5 What the validation covers

The release validation comprises a static source audit and an acceptance validator. Between them they check: declaration parity between source and registry; uniqueness of every public name; that the axiom-receipt surface covers exactly the public theorem surface; that every theorem’s recorded axiom classification agrees with its own receipt; import-graph acyclicity; registry closure over candidates, objects, targets, no-gos, corollaries and decisions; the finite census; the immutability of inherited scaffolds; and a delivery hash ledger over the complete artefact set.

15.6 The regeneration discipline

The registry-derived tables of this paper were generated from the recorded assets and pasted into the manuscript, not transcribed by hand. Exactly one generator is part of the release: it emits the disposition census, axiom receipt, Phase-5 census and no-go tables and the declaration and candidate columns of the corollary table (Section A, Section B) as paste-in files. The corollary table’s provenance column and the tables of Section D were generated from the same registries by the pre-ship audit’s own tooling, which is not part of the release; the assay tables of Section 10 and the per-entry reference of Section C were generated from the registries during drafting and pasted in. A number typed into prose can drift from the artefact it describes, and the drift is silent: nothing fails, the paper simply becomes false in a small way that compounds.

What this buys is that each table has a registry it can be compared against, so a disagreement is found by regenerating and comparing rather than by someone noticing. What it does not buy

should be stated as plainly: no table is regenerated by the release pipeline, the manuscript inputs are not rebuilt when an asset changes, and the release validators check the assets against each other, not the manuscript against the assets. Every figure printed here was checked against the registries in the pre-ship audit.

It also does not guarantee that the source is right, that the figure is the relevant one, or that the prose around it draws the correct conclusion. A generated table can be accurate and misleadingly framed.

15.7 What the validation does not cover

It does not check that the definitions are the right definitions, that the laws are interesting, or that the finite carriers are the relevant ones. Those are mathematical and editorial judgements, and no validator substitutes for them. The checks establish internal consistency and reproducibility; the argument for the content is the paper.

16 Named negative and regression theorem surface

The negative results are collected here as a surface in their own right, because a negative result quoted without its scope is the most likely way this work gets misused.

16.1 The eleven no-go fronts

Each front is a scoped obstruction with named escape routes. The escapes are not caveats appended for safety; they are part of the result, and a front stated without them is a different and stronger claim than the one established.

Each front is given below with its declared scope and its named escape theorems. The escapes are in the body rather than in an appendix deliberately: an obstruction whose escapes are one page away is quoted without them.

NGVII-01 — No first extension without seed or reachable generator. *Scope:* a declared closed admission regime and a declared permitted-first-extension family. *Escapes:* an admitted seed; a reachable generator; open external provision. This is the only front with three escapes: the first two are the two ways a closed family can lawfully acquire a first move, and the third escapes by the family not being closed. Formalised as Theorem 5.4. *Finite controls:* failure scenario TTW-S01; positive control TTW-S02 (Section D for the countermodel data). *Provenance:* RECOVERED SBT (E1 / F12); SHARPENED (three exact escapes). Foundations V E1 (Thm. 5.1) [34]; Foundations IV F12 (Thm. 3.4) [33]. Same source as VII-C003. *Lean:* FoundationsVII.NoGo.NGVII_01_no_first_extension_without_seed_or_generator, FoundationsVII.NoGo.NGVII_01_escape_admitted_seed, FoundationsVII.NoGo.NGVII_01_escape_reachable_generator, FoundationsVII.NoGo.NGVII_01_escape_open_external_provision; receipt axioms: none.

NGVII-02 — No resemblance-only independence-sensitive join. *Scope:* independence-sensitive credit in the declared resemblance, contact and source profile. *Escape:* witnessed independent contact. Resemblance between two theories is not contact, and contact that is not witnessed as independent does not support an independence-sensitive claim. *Finite controls:* failure scenarios TTW-S05, TTW-S15; positive control TTW-S08 (Section D for the countermodel data). *Provenance:* SHARPENED (resemblance is not witnessed independent contact). Foundations IV F49 (Thm. 8.16) [33]. Differently scoped from the common-source law. *Lean:*

FoundationsVII.NoGo.NGVII_02_no_resemblance_only_independence_join, FoundationsVII.NoGo.NGVII_02_escape_witnessed_independent_contact; receipt axioms: none.

NGVII-03 — No retrospective self-certification. *Scope:* an explicit event-precedence relation; no global clock is assumed. *Escape:* a prospective certificate. The scope clause matters more than it looks: the front does not require a universal time, only a recorded precedence between the certificate and the evidence it is about. *Finite controls:* failure scenario TTW-S04; positive control TTW-S03 (Section D for the countermodel data). *Provenance:* RECOVERED SBT (pre-outcome credit; protocol trap); SHARPENED (event-precedence without a global clock). Foundations V E10.1 (Cor. 6.7) [34]; Foundations I T-AOT-02 (Thm. 5) [30]. Differently scoped: stated over an explicit precedence relation. *Lean:* FoundationsVII.NoGo.NGVII_03_no_retrospective_self_certification, FoundationsVII.NoGo.NGVII_03_escape_prospective_certificate; receipt axioms: none.

NGVII-04 — No automatic total-lens transfer. *Scope:* a total source lens transported to a partial, self-owned target without a certified adapter. *Escape:* a certified adapter. Totality does not survive transport by itself; the adapter is what carries it, and the adapter must be certified rather than assumed. *Finite controls:* failure scenario TTW-S23; positive control TTW-S08 (Section D for the countermodel data). *Provenance:* RECOVERED SBT (information-loss and recoverability discipline); SHARPENED (a certified adapter as the only route for totality across a partial target). Foundations IV F34 (Thm. 8.3) [33]. Differently scoped. *Lean:* FoundationsVII.NoGo.NGVII_04_no_automatic_total_lens_transfer, FoundationsVII.NoGo.NGVII_04_escape_certified_adapter; receipt axioms: none.

NGVII-05 — No unpriced observer-native formation credit. *Scope:* native or endogenous credit where positive occupancy exceeds charged occupancy. *Escapes:* an external observer; zero occupancy. Formalised as Theorem 6.9. The second escape is not a loophole — an observer that genuinely occupies nothing owes nothing — but it must be certified, not asserted. *Finite controls:* failure scenario TTW-S22; positive control TTW-S02 (Section D for the countermodel data). *Provenance:* RECOVERED SBT (no-overreading of suppressed content; priced exposure); SHARPENED (unpriced occupancy refusal with two escapes). Foundations III Thm. 23 [32]; Foundations V E6/E9 [34]. Same source as VII-C029. *Lean:* FoundationsVII.NoGo.NGVII_05_no_unpriced_observer_native_credit, FoundationsVII.NoGo.NGVII_05_escape_external_observer, FoundationsVII.NoGo.NGVII_05_escape_zero_occupancy; receipt axioms: none.

NGVII-06 — No join from contact alone. *Scope:* a witnessed contact profile lacking a composite, retention, and anti-product novelty. *Escape:* a strict-join certificate meeting all three conjuncts of Theorem 6.4. This is the front that keeps witnessed contact strictly below a join: contact is real, and it is not yet a join. *Finite controls:* failure scenario TTW-S06; positive control TTW-S08 (Section D for the countermodel data). *Provenance:* RECOVERED SBT (strict extension); SHARPENED (contact alone is below a join). Foundations IV F8 (Lemma 4.3) [33]; Foundations III Thm. 12 [32]. Same source as VII-C009. *Lean:* FoundationsVII.NoGo.NGVII_06_no_join_from_contact_alone, FoundationsVII.NoGo.NGVII_06_escape_complete_strict_join_certificate; receipt axioms: none.

NGVII-07 — No product or common-refinement strictness credit. *Scope:* the declared comparison baseline, where a mere product or common refinement still factors through that baseline. *Escape:* an anti-product witness. This front is why Theorem 6.4’s third conjunct excludes factorization through the product and the common refinement as well as through either parent.

Finite controls: failure scenarios TTW-S07, TTW-S14, TTW-S15; positive control TTW-S08 (Section D for the countermodel data). *Provenance*: RECOVERED SBT (unification as common refinement); SHARPENED (anti-product witness required against the declared baseline). Foundations IV F51 (Thm. 8.18) [33]. Differently scoped. *Lean*: FoundationsVII.NoGo.NGVII_07_no_product_common_refinement_strictness_credit, FoundationsVII.NoGo.NGVII_07_escape_anti_product_witness; receipt axioms: none.

NGVII-08 — No one-lineage source-independence credit. *Scope*: independence-sensitive source claims resting on a single uncertified lineage. *Escape*: a disjoint-root certificate. *Finite controls*: failure scenarios TTW-S05, TTW-S11; positive control TTW-S08 (Section D for the countermodel data). *Provenance*: RECOVERED SBT (carried source exclusion); SHARPENED (disjoint-root certificate). Foundations V D3 (Prop. 3.9) [34]. Same source as VII-C010. *Lean*: FoundationsVII.NoGo.NGVII_08_no_one_lineage_source_independence_credit, FoundationsVII.NoGo.NGVII_08_escape_disjoint_root_certificate; receipt axioms: none.

NGVII-09 — No positive-cost join credit without payment. *Scope*: positive-cost join credit in the declared payment and observer-pricing profile. *Escapes*: a paid channel; a certified zero-cost channel. Formalised within Theorem 6.10. A zero-cost channel is permitted; an uncertified *claim* of zero cost is not. *Finite controls*: failure scenario TTW-S10; positive control TTW-S08 (Section D for the countermodel data). *Provenance*: RECOVERED SBT (no free distinction; priced access); SHARPENED (paid or certified zero-cost channel). Foundations IV F12 (Thm. 3.4) [33]; Foundations V E6 (Thm. 6.1) [34]. Same source as VII-C011, VII-C035. *Lean*: FoundationsVII.NoGo.NGVII_09_no_positive_cost_join_credit_without_payment, FoundationsVII.NoGo.NGVII_09_escape_paid_channel, FoundationsVII.NoGo.NGVII_09_escape_certified_zero_cost_channel; receipt axioms: none.

NGVII-10 — No arrow from holonomy alone. *Scope*: the declared holonomy, drive, path-asymmetry, reversal, budget and audit profile. *Escape*: an independently driven arrow. Formalised as Theorem 8.3; the escape is precisely the drive certificate the theorem requires. *Finite controls*: failure scenario TTW-S17; positive control TTW-S18 (Section D for the countermodel data). *Provenance*: RECOVERED SBT (no arrow from protocol holonomy alone); SHARPENED (typed interaction-profile statement). Foundations I T-AOT-02 (Thm. 5) [30]; NG_PROTOCOL_TRAP (Thm. 3.3) [38]; §7 [39]. Weaker and differently scoped: no KL statement is reproved. *Lean*: FoundationsVII.NoGo.NGVII_10_no_arrow_from_holonomy_alone, FoundationsVII.NoGo.NGVII_10_escape_independent_driven_arrow; receipt axioms: none.

NGVII-11 — No occurrence from reachability alone. *Scope*: operational profiles satisfying the declared execution, reachability and firing implication spine. *Escape*: a fired occurrence. Formalised as Theorem 7.7. *Finite controls*: failure scenario TTW-S20; positive control TTW-S21 (Section D for the countermodel data). *Provenance*: SHARPENED (reachability is not occurrence, over the declared operational spine). Foundations V E5 (Thm. 5.5) reachability records [34]. Differently scoped. *Lean*: FoundationsVII.NoGo.NGVII_11_no_occurrence_from_reachability_alone, FoundationsVII.NoGo.NGVII_11_escape_fired_occurrence; receipt axioms: none.

Reading the escapes. Every front has at least one, and that is structural rather than accidental. A front with no escape would not be a scoped obstruction but a universal impossibility claim, and this framework does not make those: each front says that a specific credit cannot be taken *through*

a *specific route*, and names the routes that remain open. The finite assays enumerate both sides, so the boundary between an obstruction and its escapes is exhibited rather than asserted.

16.2 The twenty-seven named countermodels

Each countermodel refutes a *named* claim and is rejected for a *stated* reason. The naming discipline matters: a countermodel that refutes “the idea that theories combine freely” refutes nothing checkable, whereas one that refutes “every witnessed contact yields a composite object” can be inspected, reproduced, and disputed.

Among them: a shared carrier without lawful peer transport; lawful transport without a composite object; common origin without shared access; strictness without direction; a join that dissolves a parent obstruction while creating a new one at the composite interface; and endogenous formation certified while an executor is hidden.

16.3 Two refutations

Two candidates entered the programme as proposed laws and left refuted. Both are reported with what survives, because a refutation that discards the conditional content is as misleading as an overclaim.

Refinement monotonicity (Remark 6.5). The unconditional claim is false: CM-22 (TTW-S24, REFINEMENT_DESTROY_JOIN). The four-way typed classification — preserve, strengthen, weaken, destroy — stands, and all four classes are populated over the declared carrier (P3-E07, 576 raw, 138 accepted).

Universal conserved contact degree (Remark 8.6). The claim that contact admits a universal conserved scalar degree is false over the declared assay (P3-E11, 2,880 raw, 780 accepted; CM-21, CM-25). Typed ledgers remain controlling, and the local accounting discipline of Definition 6.8 is unaffected.

16.4 Regression surface

Every negative result above is accompanied by a frozen fixture with a recorded expected status. A change to the development that causes a no-go to stop firing, or to start firing where it should not, fails the fixture rather than passing silently. This is what makes the negative surface maintainable: without it, a refactor that weakens an obstruction is indistinguishable from one that does not touch it.

A Catalog tables

The tables in this appendix were generated from the release registries rather than transcribed: the disposition census, the no-go table and the declaration and candidate columns of the corollary table by the release’s exhibit generator, the corollary table’s provenance column by the pre-ship audit’s tooling from the same registries. A figure that disagrees with its registry is found by regenerating and comparing (Section 15).

A.1 Terminal disposition census

A.2 Candidate closure

Table 1. Terminal disposition census. Kernel verification does not alter disposition.

Disposition	Count
Conditional Theorem	15
Formal Schema	15
Refuted Candidate	2
Closed Deferral	2
Constructive Countermodel	1
Lean Decidable Finite	1
Lean Kernel Proved	0

Table 2. No-go fronts with their named escape routes. A front is never cited without these.

Front	Candidates	Named escape routes
FVII-SCI02-NGVII-01	VII-C003	NGVII_01_escape_admitted_seed NGVII_01_escape_reachable_generator NGVII_01_escape_open_external_provision
FVII-SCI03-NGVII-02	VII-C010, VII-C035	NGVII_02_escape_witnessed_independent_contact
FVII-SCI02-NGVII-03	VII-C006	NGVII_03_escape_prospective_certificate
FVII-SCI02-NGVII-04	VII-C005	NGVII_04_escape_certified_adapter
FVII-SCI02-NGVII-05	VII-C029	NGVII_05_escape_external_observer NGVII_05_escape_zero_occupancy
FVII-SCI03-NGVII-06	VII-C007, VII-C009, VII-C035	NGVII_06_escape_complete_strict_join_certificate
FVII-SCI03-NGVII-07	VII-C009, VII-C026, VII-C035	NGVII_07_escape_anti_product_witness
FVII-SCI03-NGVII-08	VII-C010, VII-C035	NGVII_08_escape_disjoint_root_certificate
FVII-SCI03-NGVII-09	VII-C011, VII-C035	NGVII_09_escape_paid_channel NGVII_09_escape_certified_zero_cost_channel
FVII-SCI04-NGVII-10	VII-C017, VII-C032	NGVII_10_escape_independent_driven_arrow
FVII-SCI02-NGVII-11	VII-C021	NGVII_11_escape_fired_occurrence

A.3 Cross-family corollaries

Table 3. Cross-family corollaries, supporting candidates and provenance. Every corollary is SHARPENED: a typed record stating together facts inherited from the entries it spans; canonical sources are those entries’ (Section D).

Corollary	Declaration	Candidates	Provenance
FVII-COR-001	certified_has_temporal_source_join_and_payment	VII-C002,VII-C006,VII-C009,VII-C011	SHARPENED; spans C002, C006, C009, C011; axioms none
FVII-COR-002	retrospective_registration_blocks_prospective_join_credit	VII-C002,VII-C006,VII-C009,VII-C011	SHARPENED; spans C002, C006, C009, C011; axioms none
FVII-COR-003	certified_preserves_declared_source_alignment	VII-C002,VII-C006,VII-C009,VII-C011	SHARPENED; spans C002, C006, C009, C011; axioms none
FVII-COR-004	certified_has_independence_payment_and_capacity_bound	VII-C009,VII-C010,VII-C011,VII-C029	SHARPENED; spans C009, C010, C011, C029; axioms propext
FVII-COR-005	source_and_payment_gates_remain_distinct	VII-C009,VII-C010,VII-C011,VII-C029	SHARPENED; spans C009, C010, C011, C029; axioms none
FVII-COR-006	certified_noninteraction_excludes_strict_join	VII-C008,VII-C033	SHARPENED; spans C008, C033; axioms none
FVII-COR-007	certified_noninteraction_retains_escape_routes	VII-C008,VII-C033	SHARPENED; spans C008, C033; axioms none
FVII-COR-008	certified_accumulates_resources_and_preserves_residuals	VII-C019,VII-C027,VII-C036	SHARPENED; spans C019, C027, C036; axioms propext
FVII-COR-009	join_created_needle_blocks_zero_residual_reading	VII-C019,VII-C027,VII-C036	SHARPENED; spans C019, C027, C036; axioms none
FVII-COR-010	preserving_refinement_with_valid_descent_preserves_three_gates	VII-C015,VII-C030,VII-C034	SHARPENED; spans C015, C030, C034; axioms none
FVII-COR-011	destroying_refinement_exhibits_a_failed_join_gate	VII-C015,VII-C030,VII-C034	SHARPENED; spans C015, C030, C034; axioms none
FVII-COR-012	valid_descent_does_not_erase_refinement_obstruction	VII-C015,VII-C030,VII-C034	SHARPENED; spans C015, C030, C034; axioms none
FVII-COR-013	no_free_first_join_two_gate_obstruction	VII-C003,VII-C011,VII-C035	SHARPENED; spans C003, C011, C035; axioms none
FVII-COR-014	seeded_and_paid_first_join_escape	VII-C003,VII-C011,VII-C035	SHARPENED; spans C003, C011, C035; axioms none
FVII-COR-015	strict_join_and_holonomy_do_not_replace_drive	VII-C009,VII-C017,VII-C032	SHARPENED; spans C009, C017, C032; axioms none
FVII-COR-016	holonomy_with_independent_drive_supports_arrow	VII-C009,VII-C017,VII-C032	SHARPENED; spans C009, C017, C032; axioms none
FVII-COR-017	certified_excludes_contact_and_occurrence_at_declared_scope	VII-C021,VII-C023,VII-C033	SHARPENED; spans C021, C023, C033; axioms none
FVII-COR-018	point_null_plus_bounded_horizon_is_not_automatically_global	VII-C021,VII-C023,VII-C033	SHARPENED; spans C021, C023, C033; axioms none
FVII-COR-019	certified_requires_pricing_and_internal_execution	VII-C013,VII-C029	SHARPENED; spans C013, C029; axioms none
FVII-COR-020	unpriced_observer_blocks_joint_native_endogenous_credit	VII-C013,VII-C029	SHARPENED; spans C013, C029; axioms none
FVII-COR-021	zero_occupancy_escape_preserves_endogenous_credit	VII-C013,VII-C029	SHARPENED; spans C013, C029; axioms none

Table 4. Executed `#print axioms` census over all 732 public theorems.

Axiom dependency	Theorems	Share (%)
no axioms	625	85.4
propext	97	13.3
propext, Quot.sound	10	1.4

Table 5. Phase-5 closure-assay census (eleven assays). This is the census the release validation checks directly; the all-assay totals are in Section 2.5.

Carrier partition	Cases
Raw	86912
Canonical	84864
Accepted	21081
Rejected	63783

B Lean surface and finite census

B.1 Executed axiom receipt

The census below is computed from the executed `#print axioms` output, per theorem. Per Remark 14.1 it must not be summarized as “732 axiom-free theorems”.

B.2 Bounded-carrier census

Finite evidence in this paper is exhaustive over its declared carriers and over nothing else (Remark 10.1); the eleven Phase-5 closure assays are tabulated here, and the 43-assay breakdown is in Section 10.

B.3 Build and reproduction

The development is built on `leanprover/lean4:v4.28.0` with no external dependency. The full pipeline — baseline regeneration, clean kernel build, five finite runners, five cross-implementation differentials, axiom capture, and re-validation — is a single recorded command, described in Section 14.

C Per-entry reference

One entry per catalog candidate, generated from the release registry. Each gives the disposition, the conclusion as recorded in the terminal registry, the nonclaims that travel with it, and the declared escape routes where it is an obstruction. For seven entries (VII-C005, VII-C015, VII-C017, VII-C025, VII-C029, VII-C032, VII-C034) the Step-3 conclusion sentence claimed more than the kernel-replayed declarations establish; the terminal registry now carries the Lean-strength sentence printed here, and retains the Step-3 wording in a separate historical field. The registry conclusion and the body statement therefore agree; the body is where the claims are argued.

VII-C001 — Accessible-domain state normal form

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-02

Conclusion. A theory package admits a typed operational-domain record separating expressible, present, exposed, recoverable, admissible, reachable, and occurrent statuses; converses require explicit witnesses. **Nonclaims.**

- Domain width is not capability.
- No total order among all status coordinates is claimed.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E11; countermodels CM-08, CM-09, CM-12; 48 Lean declarations.

VII-C002 — Lawful admission transition system

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-02

Conclusion. Admission, expiry, revocation, retraction, and rollback are typed transitions whose execution requires source, budget, guard, and audit witnesses distinct from rule text. **Nonclaims.**

- A well-formed rule need not be executable.
- A reachable transition need not fire.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E01; countermodels CM-08, CM-09, CM-18; 41 Lean declarations.

VII-C003 — Bootstrap obstruction theorem

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-02

Conclusion. A closed admission regime with no admitted seed and no reachable generator has no lawful first extension. **Nonclaims.**

- External provision is not forbidden.
- The theorem is scoped to the declared transition family.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E06; countermodels CM-19, CM-20; 31 Lean declarations.

VII-C004 — Neutral seed and provisioning certificate

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-02

Conclusion. A seed receives neutral-provisioning status only when its source, symmetry, task-blindness, preregistration, and control coordinates are recorded before use. **Nonclaims.**

- Neutrality is audit-relative, not metaphysical independence.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. countermodels CM-01, CM-19; 44 Lean declarations.

VII-C005 — Common-origin non-transfer law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-02

Conclusion. Common origin, common carrier and common instrument, even jointly, do not entail shared access; shared access does not entail source independence; and a shared lineage defeats rather than supplies independence-sensitive join credit. **Nonclaims.**

- Common origin may still be relevant evidence when correctly typed.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. countermodels CM-01, CM-12, CM-26; 35 Lean declarations.

VII-C006 — Prospective commitment certificate

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-02

Conclusion. A prospective commitment is a timestamped, preregistered, finite-budget rule that fixes later admission conditions before the target evidence is observed. **Nonclaims.**

- Commitment does not guarantee reachability or success.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E01; countermodels CM-20, CM-09; 26 Lean declarations.

VII-C007 — Join-entry record normal form

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-03

Conclusion. Entry to the join calculus requires parent packages, typed contact surface, witnessed contact or pending status, source and licensing ledgers, budget, obstruction semantics, and an append-only audit record. **Nonclaims.**

- Record completeness does not prove a join exists.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E01, P3-E02; countermodels CM-02, CM-03, CM-14; 32 Lean declarations.

VII-C008 — Join existence and obstruction status calculus

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-03

Conclusion. Within a declared finite interface family, join attempts are classified as no contact, contact, common refinement, composite, strict join, typed obstruction, or coverage-qualified non-interaction. **Nonclaims.**

- The taxonomy is not universal without a completeness theorem.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E02, P5-E03, P5-E11; countermodels CM-02, CM-13, CM-27; 40 Lean declarations.

VII-C009 — Strict join certificate and anti-product witness

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-03

Conclusion. A strict join requires composite objecthood, recoverable parent retention, and a distinction or operation that does not factor through either parent or their declared mere product/common refinement. **Nonclaims.**

- Strict join does not imply a P6 arrow.
- Parent strictness alone does not prove joint strictness.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E03, P3-E10, P5-E01, P5-E02, P5-E07; countermodels CM-03, CM-13, CM-14, CM-16; 44 Lean declarations.

VII-C010 — Source-independence gate

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-03

Conclusion. Independence-sensitive join claims require a source/ancestry witness separate from behavioral nonfactorization; native and bridged records retain their source types. **Nonclaims.**

- Source independence is not sufficient for strict novelty or causal independence.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.

- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E04, P5-E02; countermodels CM-01, CM-26; 28 Lean declarations.

VII-C011 — Join budget and payment ledger

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-03

Conclusion. A join ledger prices interface access, parent retention, joint novelty, observer/instrument occupancy, cross-costs, and explicit refunds; zero total cost requires a certified zero-cost channel. **Nonclaims.**

- No universal conserved interaction currency is asserted.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E05, P3-E06, P5-E01, P5-E02, P5-E06; countermodels CM-11, CM-21; 32 Lean declarations.

VII-C012 — Enablement record and attribution calculus

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-04

Conclusion. Enablement records identify the operation made available, its source, execution witness, budget, target effect, and attribution to theorist, carrier, peer theory, observer, environment, endogenous system, or mixed source. **Nonclaims.**

- Enablement does not imply descent or sufficient determination.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E01, P5-E11; countermodels CM-06, CM-07, CM-17; 39 Lean declarations.

VII-C013 — Endogenous enablement criterion

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-04

Conclusion. Enablement is endogenous only when the system carries, reaches, executes, and audits the enabling generator within the declared boundary and pays its resource costs. **Nonclaims.**

- Endogenous does not mean uncaused or environmentally isolated.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E02, P5-E09; countermodels CM-11, CM-17; 35 Lean declarations.

VII-C014 — Birth/contact-surface classification

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-04

Conclusion. Layer birth is classified by who performs closure, what new object survives, whether participants are created or only related, and whether the birth occurs at or through a contact surface. **Nonclaims.**

- Contact does not itself constitute birth.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E03; countermodels CM-02, CM-17; 42 Lean declarations.

VII-C015 — Typed transmission and descent-fidelity law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-04

Conclusion. Upward, downward, and peer transmission are typed transmission records with explicit payload, loss, ambiguity, and source fields; a pure downward selection (no external insertion, lower-fact provenance preserved) cannot create lower-carrier facts absent before the selection. **Nonclaims.**

- Construction direction is not automatically temporal directionality.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E04, P5-E05; countermodels CM-06, CM-22, CM-23; 46 Lean declarations.

VII-C016 — Peer contact and transport without join

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-03

Conclusion. Certified content may become available to a peer through a typed contact/transport record without forming a joint theory; destination ownership and provenance remain explicit. **Nonclaims.**

- Peer transport is not joint objecthood or strict novelty.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E01; countermodels CM-02, CM-19; 42 Lean declarations.

VII-C017 — Interaction-order residue and holonomy

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-04

Conclusion. Two legal route results on the same typed target and members that differ in order

or bracketing and disagree in predictive value define a nonzero route residue; nonzero residue is interaction holonomy, not by itself directionality. **Nonclaims.**

- P3 holonomy does not imply a P6 arrow.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E09, P4-E10, P5-E07, P5-E10; countermodels CM-10, CM-24; 46 Lean declarations.

VII-C018 — Admission confluence and seed-dependence law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-04

Conclusion. Distinct lawful admission/refinement routes are confluent exactly under declared critical-pair and audit-equivalence conditions; otherwise fixed points may depend on route or initial seed partition. **Nonclaims.**

- One commuting square does not prove global confluence.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E07, P4-E08, P5-E10; countermodels CM-23, CM-24; 39 Lean declarations.

VII-C019 — Access/join residual and obstruction-dissolution ledger

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-03

Conclusion. Inaccessible, unadmitted, expired, failed, or join-created distinctions remain typed residuals; ascent, peer join, refinement, or declared termination may discharge them, with cross-terms recorded explicitly. **Nonclaims.**

- Access growth need not monotonically reduce all obstructions.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E08, P5-E04, P5-E11; countermodels CM-22, CM-25; 25 Lean declarations.

VII-C020 — Bridge and semantic-withdrawal discipline

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-01

Conclusion. Every imported law or interaction claim carries source/target maps, hypotheses, audit path, nonclaims, and append-only failure or withdrawal records; citations alone never license transport. **Nonclaims.**

- A bridge certificate does not strengthen its source theorem.

- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. countermodels CM-12, CM-18; 10 Lean declarations.

VII-C021 — Reachability, guard activity, and horizon law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-02

Conclusion. Lawfulness, non-vacuity, reachability, firing, and occurrence are separate; finite-horizon non-occurrence licenses only horizon-qualified conclusions unless reachability and guard power are independently established. **Nonclaims.**

- Non-occurrence alone is not impossibility.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E08; countermodels CM-08, CM-09, CM-18; 45 Lean declarations.

VII-C022 — Exposure, recoverability, admissibility, and rigidity calculus

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-02

Conclusion. Determination, exposure, recoverability, and admissibility are independent interface coordinates; exposure overread and spurious rigidity occur when distinctions or constraints are supplied by operations absent from the substrate. **Nonclaims.**

- Exposure is not determination; recoverability is not permission.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. countermodels CM-03, CM-12; 48 Lean declarations.

VII-C023 — Negative-result quantifier discipline

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-01

Conclusion. The force of a negative result is fixed by its quantifiers and search-family closure: a point null, bounded search, exhaustive finite family, and theorem-grade impossibility have distinct conclusions. **Nonclaims.**

- A failed instance never refutes an unrestricted existential claim.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.

- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
- Evidence.** assays P5-E08; countermodels CM-18, CM-27; 5 Lean declarations.

VII-C024 — Claim-grade and normative-specification discipline

Disposition: FORMAL SCHEMA **Landed:** FVII-SCI-01

Conclusion. VII statements preserve theorem/schema/calibration/conjecture/philosophy grades, label necessary and sufficient conditions explicitly, and are judged against normative specifications rather than illustrative examples. **Nonclaims.**

- The readiness dossier is not a proof or paper draft.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. countermodels CM-12, CM-18; 10 Lean declarations.

VII-C025 — Two-Theory World detector contract

Disposition: LEAN DECIDABLE FINITE **Landed:** FVII-SCI-01

Conclusion. In the finite reference world a detector contract records, for the interaction laws, append-only evidence with non-empty signal, null and falsifier case lists, non-empty same-source, no-contact, scheduling and relabeling control lists, a false-positive cost strictly above the false-negative cost, and the frozen scenario and countermodel identifiers; it is well formed and complete over the 24 scenarios and 27 countermodels. **Nonclaims.**

- Passing finite scenarios does not prove universal laws.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E11; countermodels CM-01, CM-02, CM-03, CM-04, CM-05, CM-06; 8 Lean declarations.

VII-C026 — Categorical reduction decision

Disposition: CLOSED DEFERRAL **Landed:** FVII-SCI-03

Conclusion. Reduction of the VII join to product, pullback, pushout, or another categorical construction remains conditional on an explicit category of packages, interfaces, and admissible morphisms. **Nonclaims.**

- No unconditional categorical reduction is claimed in Step 3.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E10; countermodels CM-13, CM-27; 21 Lean declarations.

VII-C027 — Enablement-chain composition law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-04

Conclusion. Enablement relations compose only when source types, execution witnesses, budgets, and residual obligations compose; chain order may carry holonomy or accumulated debt. **Nonclaims.**

- Enablement is not assumed transitive without hypotheses.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E06, P5-E04; countermodels CM-07, CM-24; 45 Lean declarations.

VII-C028 — Primitive-operation algebra readiness criterion

Disposition: CLOSED DEFERRAL **Landed:** FVII-SCI-04

Conclusion. A generators-and-relations algebra is deferred until primitive operations, equivalences, composition domains, and categorical semantics are fixed and proved nonredundant. **Nonclaims.**

- No complete primitive algebra is claimed.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E12; countermodels CM-24; 26 Lean declarations.

VII-C029 — Observer and instrument occupancy law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-02

Conclusion. An observer or instrument using bounded shared capacity cannot certify native/endogenous formation while its occupancy is under-charged (charged occupancy below actual occupancy) or while its source role or cost is left off the system ledger. **Nonclaims.**

- Observation is not prohibited; unrecorded load-bearing observation is.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P5-E02, P5-E09; countermodels CM-11, CM-21; 29 Lean declarations.

VII-C030 — Parent refinement, retention, and join descent

Disposition: REFUTED CANDIDATE **Landed:** FVII-SCI-03

Conclusion. A join certificate records which parent objects, laws, and source identities remain recoverable; parent refinement may preserve, strengthen, weaken, or destroy the join depending on compatibility and strictness witnesses. **Nonclaims.**

- Retention alone is not strict novelty.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E07, P5-E05; countermodels CM-05, CM-22; 29 Lean declarations.

VII-C031 — Contact-degree conservation decision

Disposition: REFUTED CANDIDATE **Landed:** FVII-SCI-03

Conclusion. No conserved contact degree is assumed; candidate quantities must first be defined, shown invariant under declared joins, and separated from bookkeeping artifacts. **Nonclaims.**

- No conserved contact quantity is claimed.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E11; countermodels CM-21, CM-25; 22 Lean declarations.

VII-C032 — Interaction arrow, irreversibility, and cross-time contact

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-04

Conclusion. Arrow or irreversibility claims require a drive certificate recording drive, path asymmetry, failed reversal, budget and audit, distinct from order residue, with null-versus-driven controls exhibited alongside it; cross-time contact uses explicit synchronization or partial-order witnesses rather than assumed simultaneity. **Nonclaims.**

- Order sensitivity and entropy production internal to a parent do not alone prove an inter-theory arrow.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E10, P4-E11, P5-E07, P5-E10; countermodels CM-10, CM-24; 58 Lean declarations.

VII-C033 — Coverage-qualified certified non-interaction

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-03

Conclusion. Certified non-interaction is a positive, family-scoped result only when contact channels, detector power, budget, horizon, and search-family closure are recorded and all admissible cases are exhausted or theoremmatically excluded. **Nonclaims.**

- No finite null universalizes beyond its covered family.
- This Step-3 dossier is not a proof of the candidate.

- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E09, P5-E03; countermodels CM-02, CM-18, CM-27; 39 Lean declarations.

VII-C034 — Enablement without descent separation theorem

Disposition: CONSTRUCTIVE COUNTERMODEL **Landed:** FVII-SCI-04

Conclusion. There exist enablement separation profiles in which enablement holds and is recorded as load-bearing while no descent factorization of the enabled object through the enabler is recorded; enablement may also be necessary but insufficient, with alternative determinants present. **Non-claims.**

- The theorem does not deny that some enablement also descends.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P4-E05, P5-E05; countermodels CM-06, CM-07; 25 Lean declarations.

VII-C035 — No-free-join and self-bootstrap no-go family

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-03

Conclusion. A join cannot create the first capability required to execute itself, manufacture strict joint information by relabeling, or receive positive-cost join credit without a paid or certified zero-cost channel. **Nonclaims.**

- The family permits admitted seeds, declared bridges, and proven zero-cost channels.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.
- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E05, P5-E06; countermodels CM-03, CM-19, CM-21; 53 Lean declarations.

VII-C036 — Join-created obstruction and needle law

Disposition: CONDITIONAL THEOREM **Landed:** FVII-SCI-03

Conclusion. A join may dissolve parent obstructions yet create new cross-term needles localized to the composite interface; the ledger must distinguish inherited, dissolved, and newly generated residuals. **Nonclaims.**

- Joining is not monotonically obstruction-reducing.
- This Step-3 dossier is not a proof of the candidate.
- Finite reference-world success does not universalize the statement.

Escape routes.

- Narrow or change the declared carrier/interface family.

- Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Evidence. assays P3-E08, P5-E04; countermodels CM-25, CM-03; 25 Lean declarations.

D Contributions, provenance and countermodel data

D.1 Contributions table

Every catalog entry, together with its provenance labels (Section 2.4), its canonical source(s) in the earlier corpus or the external literature, and what this paper adds beyond them. The one-sentence relation to each canonical source is printed with the entry in the body. Identifiers of the earlier papers are quoted from the vendored sources; a theorem number is given only where it could be recovered from that source.

Table 6. Contributions table: ID, labels, canonical sources, what this paper adds.

ID	Labels	Canonical source(s)	What this paper adds
VII-C001	RECOVERED SBT; NEW; SHARPENED	Foundations IV Def. 2.5 (access quotient), F10 (Thm. 3.2), F11 (Thm. 3.3); Foundations II Def. 3.1 (admissible claim under honest bookkeeping); Lean adapters FT01: <code>SixBirds.ClaimRecord</code> , <code>SixBirdsFoundationsV.AccessPolicy</code>	typed seven-coordinate record; spine; six converse witnesses
VII-C002	RECOVERED SBT; SHARPENED	Foundations II Def. 3.1 (admissible claim); Foundations V E5 Reclosure Collapse (Thm. 5.5), repair-generator reachability record; Lean adapters FT01, FT02, FT17	five typed transitions; effect theorems; history survival
VII-C003	RECOVERED SBT; SHARPENED	Foundations V E1 Internalization (Thm. 5.1), <code>SixBirdsFoundationsV.NoReachableRepairGenerator</code> ; Foundations IV F12 No-Free-Distinction (Thm. 3.4), <code>SixBirdsMetaMath.FoundationsIV.Access.NoFreeDistinction.financed_refinement_retains_access</code>	regime-level statement; three escape theorems
VII-C004	RECOVERED SBT; SHARPENED	Foundations V E10.1 Intention (Cor. 6.7: positive evidence restricts action support <i>before</i> the outcome), E15 Offline Reclosure (Thm. 8.2) shared-budget record; Lean adapter FT04	typed certificate; retrospective and laundering refusals
VII-C005	RECOVERED SBT; SHARPENED	Foundations IV F49 Common-Source / Nonlocal Correlation Normal Form (Thm. 8.16), F17 Objectivity as Common Quotient (Thm. 7.1)	three proved non-transfer clauses with named witnesses
VII-C006	RECOVERED SBT; SHARPENED	Foundations V E10.1 Intention (Cor. 6.7); Foundations I T-AOT-02 protocol trap (Thm. 5): no sustained asymmetry without a genuine drive; the temporal-precedence discipline VII adopts	precedence certificate; expiry; refund theorem
VII-C007	RECOVERED SBT; SHARPENED	Foundations III §11 instrument-indexed claim records; Lean adapter FT06 <code>SixBirdsIII.InstrumentClaimRecord</code> ; Foundations V D1 repair join (Def. 3.1), contrasted, not reused	entry record; omission-attack controls

ID	Labels	Canonical source(s)	What this paper adds
VII-C008	SHARPENED; RECOVERED SBT	Foundations III §5 status and defect semantics, Thm. 5 Status-Family Separation; Foundations V E11 Institutional Rewrite (Thm. 7.1) obstruction status; Lean adapters FT09, FT10	seven-way classifier; partition theorem on the declared family
VII-C009	RECOVERED SBT; SHARPENED	Foundations IV F8 Strict Extension (Lemma 4.3), F51 Unification as Common Refinement (Thm. 8.18); Foundations III Thm. 12 Strict-Extension Nonfactorization; Foundations V E1 <code>SixBirdsFoundationsV.StrictSelfExtension</code> ; strict-test admissibility on fixed support [Strict-Test Admissibility paper]	three necessary conjuncts; anti-product witness
VII-C010	RECOVERED SBT; SHARPENED	Foundations V Def. 3.3 carried record, D3 carried-source exclusion (Prop. 3.9), <code>SixBirdsFoundationsV.CarriedSource</code> ; Foundations III source-of-truth records, §3	independence gate; one-lineage refusal
VII-C011	RECOVERED SBT; SHARPENED	Foundations V E6 Attention (Thm. 6.1), E9 Curiosity (Thm. 6.5), <code>SixBirdsFoundationsV.BudgetFeasible</code> , <code>SixBirdsFoundationsV.BindingExposureBudget</code> ; adequacy residual and blind-spot currency (Def. 4.2) [Adequacy Residuals paper]	five-kind ledger; zero-cost certification
VII-C012	RECOVERED SBT; SHARPENED	§1.3 and §11.2 (causation versus enablement) [Agents paper]; Foundations V E10 interaction status (imports the agenthood distinction)	typed record; attribution field; hidden-executor theorem
VII-C013	RECOVERED SBT; SHARPENED	Foundations V E1 Internalization (Thm. 5.1), <code>SixBirdsFoundationsV.NoReachableRepairGeneratorE</code> <code>xcludesEndogenousFamily</code> , Def. 3.3 carried record	exact criterion; hidden-executor controls
VII-C014	RECOVERED SBT; SHARPENED	canonical layer-birth classes, §2 [To Classify a Stone]	agency and participant coordinates
VII-C015	RECOVERED SBT; SHARPENED	Foundations III Thm. 30 Descent-Square Recovery, Thm. 6 Descent Through Quotient; Foundations IV F2 Descent-Repair Normal Form (Thm. 5.2); Foundations I §10.3 downward influence	typed records; no-creation law
VII-C016	RECOVERED SBT; SHARPENED	Foundations V E13 Repair Transport (Thm. 7.4), <code>SixBirdsFoundationsV.TransportTokenRecord</code> ; Foundations III §11 visibility bridges	five modes; non-entailment of composite

ID	Labels	Canonical source(s)	What this paper adds
VII-C017	RECOVERED SBT; SHARPENED	Foundations IV F3 Holonomy-Memory Repair Normal Form (Thm. 5.4); predictive witness (Def. 2.4), loop asymmetry (Thm. 3.6) [Holonomy with Memory]; Foundations I T-AOT-02 (Thm. 5)	interaction-order residue; zero-arrow witness
VII-C018	RECOVERED STANDARD; RECOVERED SBT; SHARPENED	Thm. 5.6 (elementary flatness iff local confluence) [To Flatten a Stone]; Foundations VI G8 Odometer Abelianization (Thm. 6.1)	typed record; declared-scope biconditional; seed witness
VII-C019	RECOVERED SBT; SHARPENED	Foundations IV F9 No Unstated Residual (Thm. 4.4); Foundations V E5 <code>SixBirdsFoundationsV.ResidualStatusRecord</code>	join-created class; append-only ledger
VII-C020	RECOVERED SBT; SHARPENED	Foundations III §3–4 promotion and visibility bridges, §11 bridge policy; lawful transfer, exact and defective bridge records [Needle Killer paper] (Defs. 8.1–8.3, §8); Foundations IV F19 Object Persistence (Thm. 7.3)	withdrawal ledger; citation-only refusal
VII-C021	RECOVERED SBT; SHARPENED	Foundations V E5 Reclosure Collapse (Thm. 5.5) reachability records; the scope discipline of the no-go theorems [No-Go Theorems paper]	operational spine; horizon certificate
VII-C022	RECOVERED SBT; SHARPENED	Foundations IV F11 Adequacy/No-Overread (Thm. 3.3), Def. 3.5 (hiddenness exposure), F34 Information Loss Normal Form (Thm. 8.3)	typed exposure/rigidity certificate; named overread modes
VII-C023	ELEMENTARY; RECOVERED SBT; SHARPENED	Foundations II nonclaim records; scoped no-go statements with escape routes [No-Go Theorems paper]	typed licensing record
VII-C024	RECOVERED SBT; SHARPENED	Foundations II Def. 3.1 (admissible claim), nonclaim records; Lean <code>SixBirds.AdmissibleClaim</code> , <code>SixBirds.NonclaimRecord</code>	grade-faithfulness record; no-upgrade theorems
VII-C025	RECOVERED SBT; SHARPENED	Foundations III §9 countermodel atlas; null regimes and trap checks, §4 [Agents paper]	typed contract; decidability over 51 fixtures
VII-C026	RECOVERED STANDARD; SHARPENED	Foundations IV F51 Unification as Common Refinement (Thm. 8.18)	scope repair; conditional representation
VII-C027	ELEMENTARY; RECOVERED SBT; SHARPENED	Foundations IV F3 Holonomy-Memory Repair Normal Form (Thm. 5.4); §11.2 [Agents paper]	conditional composition; order-sensitivity witness
VII-C028	ELEMENTARY; SHARPENED	none beyond the elementary monoid laws	reopen condition; monoid fragment

ID	Labels	Canonical source(s)	What this paper adds
VII-C029	RECOVERED SBT; SHARPENED	Foundations III Thm. 23 No-Overreading Suppression; Foundations V E6/E9 priced access, <code>SixBirdsFoundationsV.PositiveAccessMoveCosts</code>	occupancy pricing; two escapes
VII-C030	RECOVERED SBT; SHARPENED	Foundations IV F19 Object Persistence (Thm. 7.3), <code>SixBirdsMetaMath.FoundationsIV.StatusRecordsCoherence.ObjectPersistence.persistence_extension_retains_old</code> ; Foundations IV F2 Descent-Repair Normal Form (Thm. 5.2)	refutation; four-way classification
VII-C031	RECOVERED SBT; SHARPENED	Foundations IV F27 Conservation as Orbit Descent (Thm. 7.11)	refutation over declared assay
VII-C032	RECOVERED SBT; SHARPENED	Foundations I T-AOT-01 (Thm. 4), T-AOT-02 (Thm. 5); <code>NG_ARROW_DPI</code> (Thm. 3.1), <code>NG_PROTOCOL_TRAP</code> (Thm. 3.3) [No-Go Theorems paper]; loop asymmetry (Thm. 3.6) [Holonomy with Memory]; §7 no global time from protocol holonomy [To Notch a Stone]	typed drive certificate; cross-time witness
VII-C033	RECOVERED SBT; SHARPENED	Foundations V E11 Institutional Rewrite (Thm. 7.1), <code>SixBirdsFoundationsV.E11_NCTDObstruction</code>	coverage certificate; escape theorem
VII-C034	RECOVERED SBT; SHARPENED	§11.2 [Agents paper]; Foundations III Thm. 30 Descent-Square Recovery	constructive witnesses
VII-C035	RECOVERED SBT; SHARPENED	Foundations IV F12 No-Free-Distinction (Thm. 3.4), <code>SixBirdsMetaMath.FoundationsIV.Access.NoFreeDistinction.financed_refinement_retains_access</code> ;	three-clause family; escapes
VII-C036	RECOVERED SBT; SHARPENED	Foundations V E1 Internalization (Thm. 5.1) Foundations IV F6 No-Needles Stability (Thm. 4.1); layer-dissolving membrane theorem (Thm. 5.1) [Needle Killer paper]; Foundations VI G6 Endogenous Needle Generation	join-created needle witness

D.2 Countermodel data

Each named countermodel with the claim it refutes, the frozen scenario it is declared against, the expected status recorded in its fixture, the fixture’s SHA-256 prefix (`formalization/foundations_vii_lab/fixtures/countermodels/CM-nn.json`), and its provenance. A countermodel is SHARPENED where it is a minimized falsifier of a claim this paper types; it additionally carries RECOVERED SBT where the same separation is already exhibited in the earlier corpus.

Table 7. Countermodel data and provenance.

CM	Name	Refutes	Scenario / status	Fixture	Provenance
CM-01	Same source but no shared access	Common origin does not imply shared access or independence.	TTW-S05 INDEPENDENCE_GATE_FAILED	2cc651047d	RECOVERED SBT (Foundations IV F49, Thm. 8.16); SHARPENED (minimized)
CM-02	Common interface without witnessed contact	Compatibility does not imply contact or non-interaction.	TTW-S12 NO_EVIDENCED_CONTACT	d864055271	SHARPENED (minimized falsifier)
CM-03	Contact without strict joint information	Contact does not imply strict join.	TTW-S06 CONTACT_WITHOUT_JOIN	33a6d8ad67	RECOVERED SBT (Foundations IV F8, Lemma 4.3); SHARPENED
CM-04	Scheduling-only apparent joint novelty	Post-hoc scheduling can manufacture an apparent joint signal.	TTW-S14 FAKE_JOIN_SCHEDULING	1789701606	RECOVERED SBT (schedule trap, 27 §4); SHARPENED
CM-05	Join that loses one child	Composite formation does not imply parent retention.	TTW-S09 RETENTION_OBSTRUCTION	8a2fc25e23	RECOVERED SBT (Foundations IV F19, Thm. 7.3); SHARPENED
CM-06	Enablement without descent	An enabling condition can be load-bearing without factorization of the enabled object.	TTW-S03 PROSPECTIVE_ADMISSION	92db950645	RECOVERED SBT (27 §11.2); SHARPENED
CM-07	Enablement necessary but insufficient	An enabler may be necessary while alternatives still determine the result.	TTW-S03 PROSPECTIVE_ADMISSION	824c621b92	RECOVERED SBT (27 §11.2); SHARPENED
CM-08	Sound but unreachable admission rule	Soundness does not imply operational reachability.	TTW-S19 SOUND_UNREACHABLE	ba4260f400	SHARPENED (minimized falsifier)
CM-09	Reachable admission that never occurs	Reachability does not imply occurrence.	TTW-S20 REACHABLE_NONOCCURRENT	68e8afe940	SHARPENED (minimized falsifier)
CM-10	Holonomy with zero arrow	Route residue does not imply directionality.	TTW-S17 HOLONOMY_ZERO_ARROW	b833c4a486	RECOVERED SBT (Foundations I T-AOT-02, Thm. 5; 38 Thm. 3.3); SHARPENED
CM-11	Observer occupancy omitted from budget	Hidden observer resources defeat endogenous/native credit.	TTW-S22 UNPRICED_OBSERVER	8b165307aa	RECOVERED SBT (Foundations III Thm. 23); SHARPENED
CM-12	Total-lens result fails on partial self-owned access	Totality assumptions do not transfer silently.	TTW-S23 UNLICENSED_TOTALITY_TRANSFER	abbd587b48	RECOVERED SBT (Foundations IV F34, Thm. 8.3); SHARPENED
CM-13	Product/common refinement without strictness	Common refinement is not anti-product novelty.	TTW-S07 COMMON_REFINEMENT_NONSTRICT	2a8115eb7b	RECOVERED SBT (Foundations IV F51, Thm. 8.18); SHARPENED
CM-14	Relabeling-only fake join	Presentation change cannot manufacture strict content.	TTW-S15 FAKE_JOIN_RELABELING	a72c6d051a	RECOVERED SBT (Foundations IV F29 Presentation Invariance); SHARPENED
CM-15	Strict extension without certified objecthood	Nonfactorization does not imply closure/objecthood.	TTW-S15 FAKE_JOIN_RELABELING	8f77eef39d	RECOVERED SBT (Foundations III Thm. 12); SHARPENED

CM	Name	Refutes	Scenario / status	Fixture	Provenance
CM-16	Strict join without directionality	Strict novelty does not imply a P6 arrow.	TTW-S08 STRICT_JOIN	8bb458b3c5	RECOVERED SBT (Foundations I T-AOT-02, Thm. 5); SHARPENED
CM-17	Theorist-triggered enablement is not endogenous	Execution by the analyst fails the carried-generator condition.	TTW-S02 LAWFUL_FIRST_EXTENSION	977f5f7def	RECOVERED SBT (Foundations V E1, Thm. 5.1); SHARPENED
CM-18	Finite negative search with open family	A bounded null does not prove a universal no-go.	TTW-S19 SOUND_UNREACHABLE	db1fa94ca5	RECOVERED SBT (scope discipline of 38); SHARPENED
CM-19	Shared carrier without lawful peer transport	Co-location does not provide an executable bridge or first capability.	TTW-S01 BOOTSTRAP_BLOCKED	0659e0bf47	SHARPENED (minimized falsifier)
CM-20	Post-hoc prediction self-certification	Retrospective admission cannot earn prospective credit.	TTW-S04 RETROSPECTIVE_SELF_CERTIFICATION_REJECTED	a034cad57b	RECOVERED SBT (Foundations V E10.1, Cor. 6.7); SHARPENED
CM-21	Zero-cost join hides positive observer cost	Unpriced occupancy invalidates no-cost credit.	TTW-S22 UNPRICED_OBSERVER	8c454f88f6	RECOVERED SBT (Foundations V E6, Thm. 6.1); SHARPENED
CM-22	Parent refinement destroys an admissible join	Join admissibility need not be monotone under parent refinement.	TTW-S24 REFINEMENT_DESTROYS_JOIN	5cfdb48e1a	RECOVERED SBT (Foundations IV F19, Thm. 7.3); SHARPENED
CM-23	Seed-dependent closure fixed points	Initial partition/seed can change the terminal package.	TTW-S24 REFINEMENT_DESTROYS_JOIN	c954d623b0	RECOVERED SBT (Foundations VI G8 case (d) non-confluent witness, Thm. 6.1); SHARPENED
CM-24	Identical members assembled in different orders	Bracketing/order may change the audited result.	TTW-S16 ORDER_RESIDUE	a7e0e6257c	RECOVERED SBT (Foundations IV F3, Thm. 5.4); SHARPENED
CM-25	Join manufactures a cross-term needle	Joining need not monotonically dissolve obstructions.	TTW-S24 REFINEMENT_DESTROYS_JOIN	2fd247e91b	RECOVERED SBT (Foundations IV F6, Thm. 4.1); SHARPENED
CM-26	One-lineage pseudo-join	Behavioral difference inside one lineage does not establish source independence.	TTW-S05 INDEPENDENCE_GATE_FAILED	8ced4b2910	RECOVERED SBT (Foundations V D3, Prop. 3.9); SHARPENED
CM-27	Lawful parents with no common admissible package	Parent lawfulness does not guarantee join existence.	TTW-S13 CERTIFIED_NONINTERACTION	bed679e01f	SHARPENED (minimized falsifier)

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